On the General Science of Mathematics

ΙΑΜΒΛΙΧΟΥ ΠΕΡΙ ΤΗΣ ΚΟΙΝΗΣ ΜΑΘΗΜΑΤΙΚΗΣ ΕΠΙΣΤΗΜΗΣ

Iamblichus

translated by Claude AI

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Iamblichus of Chalcis in Coele-Syria, On the Common Mathematical Science: third discourse.

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Chapter 1

9

What the purpose of the present book is as a whole, and what the particular aims falling under it are, in how many ways and into what they are divided, and whence and from what kind of essence they derive the first causes of the inquiry appropriate to them.

9

The purpose of the present inquiry is to set forth the common theory of the mathematical sciences: what it is as a whole, and what single cause and most senior essence it has going before it; and we shall examine whether, after the single cause, there are in some way two principles of it; and after this dichotomy we shall try to reckon up, by means of a certain scientific division, whether there is any determinate number of the genera in them; and then indeed we shall examine the common forms of all the mathematical sciences by a certain common approach, not yet touching upon the theorems of each in particular. Under each of the points mentioned we shall set forth the essence in which each genus and species of mathematical objects is present; and we shall not omit to say what their contribution to the whole is and their ordering in relation to one another, what kinship they have and whence it comes to them, from what principles it is bound together, to what causes senior to itself it is led up, and how one might be able to attain them, in what respect

the study is useful and to how many goods it leads the way, and that it is choiceworthy both for itself and on account of the sciences that arise from it, and that it turns discursive thought round towards all philosophy and towards the whole science of beings and intelligibles.

10

Such in number, then, are the matters set before us to go through in this book; let us begin from the first, taking it up from the beginning.

10

In common, then, concerning all the mathematical sciences, let these axioms be laid down by us: that they are incorporeal and subsist in themselves, intermediate between the indivisible essences and those divisible about bodies, between forms and reason-principles, having been allotted the rank between the partless and the divisible, being purer than the latter and more variegated than the former; employing composition and division, yet surveying what is composed and divided in an ungenerated and eternal manner; being inferior to the intelligible essences, but prior to those in nature; surpassing visible things in beauty, order and exactness, but falling short of the intelligibles; employing likewise an intermediate commensurability and agreement; and having the power to ferry and convey across to the indivisible forms, inasmuch as they are akin to them, both leading those who become habituated to them away from bodies and turning them round towards the divine essences as by a kind of ladder leading up to the height. One must, then, view the secondary gift of the essence of incorporeals as coming down into these not from one genus of beings only, but from all the genera whatsoever that exist in that which truly is and in intellect; for from all of these there descends into the intermediate natures of the mathematical sciences the mean between the causes and the things brought to completion by them, and it joins the things that come to be to the things that are and produces a communion of them with one another.

11

Since, then, the theory of the mathematical sciences is of such scope and extends in this way to all things, mathematical science is an intermediate knowledge, exceeding intellect in composition, being of a discursive kind, comprehending many things in the same, making use rather of certain traversals and unfoldings, and of forms and reason-principles that are intermediate and not in every way limited, but which mark off the limit about the unlimited and furnish clarity in matters not very well known.

Chapter 2

What the common theory is concerning the mathematical sciences as a whole and concerning mathematical science, and whence one must grasp it and from what things one must take its defining limit, and how great an extent it has, reaching to how many common genera.

The science being of this kind, then, one must grasp it by standing apart from bodies and generation, keeping pure of imaginations and sense-perceptions, habituating oneself to things incorporeal in themselves, and making continual use of the exercise of reasonings. And it is right to set the defining limit upon them from the science of beings and the pure intellection of the pure reason-principles, the immaterial forms and the limited truth of the intelligibles; for from these one would acquire in addition the perfection and purity of the knowledge that is in them. It extends to all the genera and species of beings that are intermediate, both those that are comprehended in determinate numbers and those that proceed in a determinate manner according to certain formal differences. And some advance forward and ascend to the height, others draw near to things inferior and lower, while those that are in the middle between these join the extremes.

12

Their genera and species must be distinguished according to all these, and further they must be divided into some as existing in themselves and others as relative to other things; and their distinction must be

understood also according to the differences of quantity, and according to the divisions of the intermediate reason-principles and forms; and some of them must be posited as prior, others as posterior, according as their natures have their ordering relative to one another. One can also infer their otherness according to the cognitive faculties of the soul, all those that are intermediate and discursive, as Archytas too evidently does in the section of the cognitive line. Let such, to sketch it in outline, be the first comprehensive account of the common theory concerning the mathematical sciences; and from this point, taking the matter up again from the beginning, let us try to go through one by one the problems already stated.

Chapter 3

What the principles of the mathematical sciences as a whole are, and in what respect they differ from the other principles, all those that are principles of other essences, and how such principles furnish their cause in common to the mathematical sciences as a whole.

Let us, then, determine in common for all the mathematical sciences what the principles of mathematical essence are; for since every science comes about through its own principles, the best knowledge of mathematical essence too would be that which sets out from there. Now that the limited and the unlimited are principles of all the mathematical sciences and of all mathematical essence is clear to everyone, as the Pythagoreans hold; but each of these does not have a single account, nor the same one in every essence: rather, in the case of the intelligible forms and the immaterial reason-principles such principles are different, being intelligible in every way and immaterial and in themselves indivisible, whereas in the case of the mathematical sciences they will be causes of multitude and magnitude, of division and extension, and will partake of divisible nature and take on genera of their own, those appropriate to the mathematical sciences as a whole, and will have a share in composition and will be judged by a

discursive thinking that is other than the faculty which judges the simple, indivisible and intellective essences.

13

Some will perhaps assign motion to these principles of the mathematical sciences, namely all those who posit these principles in the soul and in the lives and powers of the soul; but it is better to place the soul in another genus of essence, and to suppose the mathematical principles and mathematical essence to be unmoved; for their forms stand always fixed, and we contemplate them in the same way and in the same respects. These principles of the unlimited and of limit, then, are of an intermediate kind, the form of limit always prevailing over that of the unlimited and bounding it within itself; hence too there is always procession towards the unlimited, but determination by that which limits. From the reason-principles that exist in intellect, then, these principles differ in that they give forth from themselves the cause of division, multitude, magnitude and composition; while from those of nature and those of soul they are separated both by being unmoved and because they belong to the intermediate incorporeals ranked in between and exist by themselves separated from matter, whereas those others also lay hold of matter. That they differ, then, from the other causes, one may be persuaded from these considerations.

14

Their community, which extends to all, must be understood from the mean conceived thus without qualification, and from the nature that is inferior to the indivisible and intelligible forms but senior to those divisible about bodies. And if one should take reason-principles, it is according to this that their community in the reason-principles must be contemplated. And indeterminacy too must likewise be posited as extending in common to all. And if certain receptacles of the mathematical forms are conceived, it is right to leave these as common to the whole multiform constitution observed in the mathematical sciences; for in this way one might comprehend their

community, which is hard to grasp in thought so as to embrace it in a single reasoning, because it is present in many and differing things, but which might with difficulty be conceived in this way. Let so much, then, be determined by us concerning the common principles as well.

Chapter 4

What the proper principles of each of the mathematical sciences are, and what distinctive property they have in themselves, what difference from one another, and what difference from all the other principles of all beings.

If one must also determine the proper principles for each of the mathematical sciences, what they are and of what sort, and what distinctive property they have in themselves, what difference from one another, and what difference from all the other principles of all beings, it is now time to go through these too. The best course of all is this: since there is a certain order among them, and some, as prior, take precedence not only in order but also in nature (for they abolish the others along with themselves but are not abolished along with them, and they bring the others in along with themselves but are not brought in along with them)[1], while others fall short in both these respects, in seniority and simplicity, for these reasons it is fitting for us too to follow their natural order, and to speak first about the first, and then in turn about the others.

15

Of mathematical numbers, then, two principles must be posited as the very first and highest: the One (which indeed one ought not yet even to call a being, because it is simple and because it is a principle of beings, and the principle is not yet such as are those things of which it is the principle), and again another principle, that of multitude, which is able of itself to furnish division as well, and for this reason we would declare it, illustrating it as fittingly as we can, to be like a wholly fluid and pliant matter; and from these, the One and the principle of multitude, the first genus is brought to completion, numbers being composed from both of these with a certain persuasive necessity. And,

going through each of the numbers one by one, one should say that it is this nature which furnishes all division to every number and, to speak generally, magnitude; whereas the fact that each of them is of a certain quality, and further determinate and one, is impressed by the undifferentiated and undivided principle setting its seal upon it. But it is perhaps not fitting to set down as evil or ugly a thing of this kind, to which it belongs to be of itself a cause of magnitude and division, and further of increase; for neither in other cases are we accustomed to place such a genus in the portion of evil, and there are times when we might perhaps truly say that the great, interwoven with a certain quality, is the cause of the magnificent and the liberal; so that it would be far from right to call it evil.

16

For if indeed one should happen to praise the nature of the One too for its self-sufficiency and for its being the cause of certain beautiful things among the numbers, how would it not be unreasonable to say that evil or the ugly is by nature receptive of such a thing? For it would then no longer turn out that evil and the ugly are in every way blameworthy, if indeed that which is receptive of something praiseworthy must itself too be called praiseworthy. Let this principle, then, be conceived by us in this way. The One, on the other hand, it is right to call neither beautiful nor good, because it is above both the beautiful and the good; for as nature proceeded further from the things at the beginning, first the beautiful appeared, and second, when the elements stand at a still greater distance, the good. The first receptacle and magnitude, then, or whatever one ought to call it, struck off the form of numbers, indeterminate in multitude, as is reasonable, but in some way determinate in form, having taken this over from the portion of the One. Now if one posits a single unlimited matter and receptacle for all things, it will in all likelihood turn out to be unreasonable that, when the form of the One comes to be in it, if indeed it is alike throughout, the same genera too should not be brought to completion once again. So the result will be that all the

genera are altogether numbers; for we shall have no fitting difference to attach, to explain why ever a nature of numbers was generated here, and after this one of lines and planes and figures, and not always the same genus, coming as it does from elements that are alike and are interwoven with one another in the same manner.

17

But if someone posits the first cause of all multitude and magnitude as one, yet as furnishing many differences within itself, on account of which it is of a nature to bring forth ever different genera throughout the whole of nature, even though the One comes to be in it alike throughout, and yet not even this always displays its own nature exactly, owing to the grossness of the matter, like a shape in some chance pieces of wood, these consequences would perhaps follow for him not unreasonably; but that the first element should have divisions into so many differences is something at which one might perhaps fittingly take offence, especially if one were to carry the point right through in accordance with these examples[2]; for what is simplest is everywhere an element. It remains, then, to posit some other cause of magnitude and, as in numbers the monad answers to the One, so to posit the point in lines; and position and the extension of places appear first in connection with lines and areas and solids, and in the same way place too makes its appearance here, owing to the fact that the difference of the receptacle hands on something proper to the genus that derives from it. Further, if one were to make this the cause of the continuous, and of what is more sullied and grosser than numbers, and say that they come from it, one would perhaps not go astray. And up to this point a second genus would have been brought to completion; for I place in the same genus lines and solids and the breadths of areas. First, then, is the matter of numbers, second that of lines and of plane and solid figures. And likewise for the other mathematical sciences, however many and of whatever kind reason may discover, their own receptacles must be presupposed.

18

Let this, then, stand so for us. The elements from which numbers come are not yet either beautiful or good; but from the composition of the One and the matter that is the cause of multitude number comes to exist, and it is in these first that being and beauty appear; next in order, from the elements of lines, geometrical essence appears, in which likewise there are being and the beautiful, and in these there is nothing either ugly or evil; but at the last, in the fourth and fifth things, those compounded from the last elements, evil arises, not in a primary way, but from certain things falling away from, and failing to keep hold of, what is according to nature.

18

From this it is evident also what difference the mathematical principles have from the others: for they stand out from the last principles because, while those are in some way corporeal, these are incorporeal; from those considered in respect of life because, while those are characterised by motion, these are unmoved; and from the intelligible principles because, while those pre-exist as indivisible, these furnish a principle of composition and division. Let the common account of the mathematical principles, and the account proper to each, be thus determined by us; and in what respect they differ from the other principles, let that be distinguished in this way.

Chapter 5

What things underlie all the mathematical sciences in common, with which lovers of learning are occupied in their study, and how it is possible to conduct the theory of them universally.

As for the things that underlie mathematical theory, those which extend in common over the whole of this science, they are all the theorems that are common, capable of applying to numbers, and capable also of applying to magnitudes, and further to harmonies and astronomy and all the rest.

19

Such are the theorems of proportions and those concerning compositions and divisions taken in common, and all that are studied concerning the equal and the unequal, in whatever condition or of whatever kind it may be, and all that examine the multiple or the divisible, or that which exceeds and that which falls short, or the determinate and the indeterminate by a common approach, or what is in itself and what is relative, or quantity simply, taking on in addition no particular species of quantity; and whatever[3] studies the order and the beauty in the mathematical forms, in so far as these are scientific, without further specifying any particular kind of beauty (for a thing of that sort already touches upon the particular sciences); and again whatever considers the fixity and stability of mathematical science, which neither changes now in one way and now in another, nor departs from its own essence, nor is conceived now thus and afterwards otherwise: this too embraces by reasoning the common things that underlie mathematical science. One must not, however, suppose that these common things are supervenient, but that they exist prior to the particulars; nor that they have their essence in the particular things and together with them, but that they possess in advance an essence senior to those and more of the nature of a principle, one which does not, however, pervade them, but is ranked before the mathematical teachings proper to each science.

20

For this very reason the knowledge of them also is common and takes precedence, and is more perfect than the knowledge of the particulars: it forms a common synoptic view of them all; it orders all the mathematical theorems from one thing and towards one thing; it looks to their kinship and their likeness to one another, and observes alongside this what is unlike and other in them; it brings together into the same and distinguishes all their primary genera and species; it studies common agreed premises and first hypotheses and definitions and theses and divisions and collections, compositions and partitions, excesses and deficiencies and applications, in whatever genera of

things having mathematical being, speaking simply and not determinately of each in particular; it discriminates the possible in these and the impossible, the necessary and the not necessary, the true and the false; and it searches out with exactness the differences in them, how many they are and of what kind.

20

Let so much be determined by us for the present, both concerning the things that underlie mathematical science in common and concerning the common mode of the theory that belongs to it.

Chapter 6

What the best use of the serious study of the mathematical sciences is, and to what end one ought to refer the best pursuit of them.

Concerning all such mathematical sciences this must be understood: that if someone grasps each of them rightly, great benefit comes to him who receives them in the proper manner; but if not, it is better always to call upon a god. The manner is this, for so much at least it is necessary to tell. Every diagram, and system of number, and every composition of harmony, and the proportion of the revolution of the stars, must be revealed to him who learns in the proper manner as being one for them all.

21

And it will be revealed, if someone learns rightly with his gaze fixed on what we are speaking of[4]; for a single natural bond of all these things will be revealed to those who think it through. But if someone goes about it in some other way, he must call upon fortune, as indeed we say; for without these no nature will ever become happy in cities, but this is the way, this the nurture, these the mathematical sciences; whether they are hard or easy, by this road one must travel. And it is not lawful to neglect the gods, now that the auspicious report of them all, told in the proper manner, has become manifest. Him who has grasped all these things in this way, him I call most truly the wisest;

for the fairest and most divine nature of all, as much of it as god has granted to human beings to behold, no one who has not beheld it will ever boast of having grasped it with ease apart from the things just now stated. In addition to these, one must bring to bear both what is taken one by one and what is taken by species[5] in each of the kinds of knowledge that belong to the mathematical sciences, until we discover the whole cosmos, which reason, the most divine of all, has set in order as visible; which the happy man first marvelled at, and then conceived a passion to learn all that is possible for mortal nature, believing that thus he will pass his life in the best and most fortunate way, and that when he has died he will come to regions befitting virtue, and that, having been initiated truly and really, and having partaken, himself being one, of a wisdom that is one, he will continue for the rest of time as a spectator of the fairest things, as many as fall to sight.

22

One must also bring together the mathematical sciences that were pursued without order in the course of education into a synoptic view of the affinity of the mathematical sciences with one another and with the nature of being; for only learning of this kind is secure in those in whom it comes to be. One must also become able, letting go of the eyes and of the rest of sense-perception, to proceed in company with truth to being itself. One must also be steadfast in the mathematical sciences, and keen, and possessed of all the other qualities that befit the best nature; since, if we bring to so great a course of learning and so great a training people sound of limb and sound of mind, and so educate them, justice herself will find no fault with us, and we shall preserve both the city and the constitution; but if we lead to these things people of another sort, we shall bring about the opposite in everything and shall pour a still greater flood of ridicule over the love of learning. And if one must tell the whole truth as it stands, in these mathematical sciences a certain organ of each person's soul is purified and rekindled, when it is being destroyed and blinded by the other

pursuits, an organ more worth saving than ten thousand eyes; for by it alone is truth seen. To those, then, who share this view, the present remarks seem extraordinarily well said; but all who have in no way perceived this will reasonably think that we are saying nothing; for they see no other benefit from them worth speaking of.

23

But this, it seems, would be no spinning of a shell, but a turning round of the soul from a day that is a kind of night to the true day, which is an ascent to being, and which we shall indeed declare to be true love of learning. We must therefore consider which of the mathematical sciences has such a power, and which mathematical science is apt to draw the soul from what comes to be towards what is. I say, then, that among the things in our sense-perceptions some do not summon intellection to an examination, since they are adequately judged by sense-perception, while others in every way urge it to examine, since sense-perception produces nothing sound; and those that do not summon are all that do not pass over at the same time into an opposite sense-perception, while those that do pass over I set down as summoning, whenever sense-perception makes plain this thing no more than its opposite, whether it strikes us from close at hand or from afar. In the following way we shall know more clearly what I mean. These, we say, would be three fingers: the smallest, the second, and the middle one. Understand me, then, as speaking of them as seen from close at hand. But consider this point about them for me. Each of them, I suppose, appears equally a finger, and in this respect at least it makes no difference whether it is seen in the middle or at the end, whether it is white or black, whether thick or thin, and everything of that sort; for in all these cases the soul of most people is not compelled to ask intellection what a finger is; for nowhere did sight itself[6] indicate at the same time that the finger is the opposite of a finger.

24

Reasonably, then, a thing of this sort would not be apt to summon intellection. But what of this? Does sight see their largeness and their smallness adequately, and does it make no difference to it whether one of them lies in the middle or at the end? And likewise touch, with thickness and thinness and hardness? And will not the other senses reveal such things defectively? Or does each of them act as follows: first, the sense that is set over the hard is compelled to be set over the soft as well, and reports to the soul that it perceives the same thing as both hard and soft? Necessarily, then, in such cases the soul in its turn is at a loss as to what the sense itself means by the hard, if indeed it says that the same thing is also soft, and as to what the sense of the light and that of the heavy mean by the light and heavy, if it indicates that the heavy is light and the light heavy; for these communications are strange to the soul and in need of examination. Reasonably, then, in such cases the soul first of all, summoning reasoning and intellection, tries to examine whether each of the things reported to it is one or two. Now if they appear to be two, each of the two appears as other and as one; if, then, each is one and both together are two, it will conceive the two as separated; for it would not conceive things unseparated[7] as two, but as one.

25

Sight too, certainly, saw large and small, as we say, yet not as separated but as something confused together; and for the sake of clarity about this, intellection in its turn was compelled to see large and small, not confused together but distinguished, in the opposite way to sight. So it is from some such point as this that it first occurs to us to ask: what, then, will the large be, and again the small? And so it was that we called the one intelligible and the other visible. This, then, is what I was attempting to say just now, that some things are apt to summon discursive thought and others are not, defining as apt to summon it those that fall upon sense-perception together with their own opposites, and those that do not as not apt to rouse intellection. What then? To which of the two classes number and the

one and the other objects of mathematical study seem to belong, it is easy to reckon from what has been said before. For if the one, or any other of the objects of mathematical study, is itself adequately seen or grasped by some other sense, it would not be apt to draw us towards essence, just as we were saying in the case of the finger; but if some contrariety is always seen together with it, so that it appears no more one than the opposite as well, there would at once be need of something to adjudicate, and the soul would be compelled in this case to be at a loss and to inquire, setting in motion the thought within itself, and to ask what the one itself is; and thus the study of the one would be among those that lead and turn one round to the vision of being. But surely the sight of it possesses this feature not least of all; for we see the same thing at once as one and as unlimited in multitude. So then, if this is true of the one, all number too is affected in the same way as it. But surely calculation and arithmetic are wholly concerned with number; and these evidently lead towards truth; pre-eminently, then, as it seems, this would be among the mathematical sciences we are seeking.

26

And the others too would in the same way be useful for science, because they lay hold of essence and are set free from becoming, and summon intellection while withdrawing us from the senses, and summon us to the vision of the nature of the things that are, and furnish the soul itself with ease in its turning round from becoming to truth and essence. One must also pursue the mathematical sciences for the sake of knowing; for in this way the soul is led powerfully somewhere upwards, and they compel people to discourse about the things themselves that are, never accepting it if someone, in discoursing with them, puts forward visible or tangible bodies; for they speak of those things which it is possible only to grasp in thought, and which it is impossible to handle in any other way. The mathematical sciences, then, are likely to be necessary, since they evidently compel the soul to employ intellection itself in pursuit of

truth itself; and moreover they make people become keener than they were, and further they afford much toil to one who learns and practises them.

26

One must also consider whether they tend in any way towards that, namely towards making it easier to behold the Idea of the Good. And all things tend that way, we say, which compel the soul to turn round to that region in which is the happiest part of being, which the soul must by every means see. So then, if a thing compels us to contemplate essence, it is fitting; if becoming, it is not fitting. And those that are pursued for the sake of knowledge are to be honoured as being mathematical sciences: all that lay hold of knowledge of what always is, and not of what at some time comes to be and perishes.

27

These, then, would be apt to draw the soul towards truth, and productive of philosophic thought, so as to direct upwards what we now wrongly keep turned downwards; for by these alone is truth seen. They must therefore be inquired into continuously and strenuously, so that it may become manifest how they stand. For in addition to everything else they also possess charm in an exceptional degree, and make the soul look upwards. Of this kind are those mathematical sciences that are concerned with being and the invisible, and all that are to be grasped by reason and discursive thought, but not by sight. One should indeed use the things that appear as models; one ought not, however, to examine them in earnest as though one were going to grasp in them the truth of equals or doubles or any other commensurability. For it is indeed absurd if someone should suppose that these things come about always in the same way and never deviate in any respect at all, though they have bodies and are visible[8], and should seek by every means to grasp the truth of them. Throughout all this one must guard against the following: that those whom we shall educate should ever attempt to learn any of these things in an incomplete form, one that does not in every case come

out at the point at which all things ought to arrive; for in this way they will be useful for the search for the beautiful and good, but pursued otherwise they are useless. And I think, too, that if the methodical inquiry into all these mathematical sciences arrives at their communion and kinship with one another, and they are reckoned together in the respect in which they are akin to one another, the pursuit of them contributes to what we want and the labour is not unprofitable; but if not, it is unprofitable.

28

For the release from the bonds and the turning round from the shadows to the images and the light, and the ascent from the underground and sensible realm to the sun and the Good, and there the continuing inability to look upon the animals and the plants and the light of the sun, that is, upon the pure forms and genera, but the ability to look upon the divine appearances in water and the shadows of the things that are, and not shadows of images cast by another light which, judged in relation to the sun, is itself of the same sort: all this pursuit of the arts which we have gone through has this power, and effects a leading up of what is best in the soul to the vision of what is most excellent among the things that are, just as then what is clearest in the body was led to the vision of what is brightest in the bodily and visible region[9]. Of some such kind is the best use of the mathematical sciences, and such is their most sovereign end.

Chapter 7

What object of science proper to it underlies each mathematical science, and how it is possible to make the common distinction of them on the basis of division, so as to know what sort of thing the one and the multitude in the mathematical sciences is, and how one ought to define it.

Since we must also determine, for each mathematical science, the proper object of science that underlies it, come, let us begin from division and distinguish the species of the objects of mathematical study with which they deal. For in this way we should most easily

learn of what sort the one and the multitude of mathematical science is, and by what differences it is distinguished. Let us begin from the following point.

The whole nature of the continuous and of the discrete is conceived along with beings, that is, with the constitution of the whole cosmos, in two ways: that of the discrete by juxtaposition and heaping together, that of the continuous by unification and mutual coherence.

29

In the strict sense, what is continuous and unified would be called magnitude, and what is juxtaposed and discrete, multitude. And according to the essence of magnitude the cosmos would be conceived and spoken of as one, and as solid and spherical, grown together with itself, extended and held together in mutual coherence; while according to the form and notion of multitude, in turn, there would be conceived the arrangement and ordering and harmony of the universe, having its constitution from, let us say, so many elements and spheres and stars and genera and animals and plants, and oppositions and likenesses. But of what is unified the cutting proceeds in every case without limit, while its increase proceeds to a determinate point; whereas of multitude, by a reciprocal relation, the increase proceeds without limit, while the cutting, conversely, proceeds to a determinate point, both being by nature and in conception unlimited, and on that account not to be circumscribed by sciences; ‘for there will not even be anything that is going to know at all, if all things are unlimited’, as Philolaus says. But since it is necessary that the nature of science be discerned in beings that have been so exactly wrought by divine providence, certain sciences, cutting off a portion of each and setting a limit to what they had embraced, called what was taken from multitude quantity, which is at once knowable, and in the same way what was taken from magnitude size.

30

And they brought both of these genera under sciences, the kinds of knowledge proper to them: quantity under arithmetic, size under geometry. But since these were not of a single form, and each of them admitted a still more particular subdivision (for of quantity one part was quantity by itself, freed from any relation to another, as for example even, odd, perfect, deficient and the like, while the other was quantity standing in some relation to something else, which is properly called relative quantity, as for example equal, unequal, multiple, superparticular, superpartient and the like; and again, of size one part exists and is conceived as at rest, the other as in motion and borne along), for this reason, reasonably enough, two other sciences came to share with the two already brought in and joined them in laying hold of the study of each object of science. For arithmetic, which received as its proper lot the inquiry into quantity by itself, was joined by music, which took as its share the systematic treatment of relative quantity (for its harmonic part and its part concerning concords profess nothing other than to articulate the relations and ratios of the notes to one another and the quantity of their excesses and deficiencies); while geometry, occupied with the examination of size that is at rest and stationary, found a helper in spherics, which was established as the judge of size in motion, that is, of the most perfect size, the one that has received an ordered and uniform motion.

31

Wherefore, since these sciences too have come to be concerned with underlying objects that are sisters, it is reasonable to consider the sciences themselves sisters as well, so that the saying of Archytas may not go unheeded[10], 'for these mathematical sciences seem to be sisters', and to hold that they cling to one another in the manner of the links of a chain and end in a single bond[11], as the most divine Plato says, and that the kinship of these mathematical sciences ought to be revealed as one to him who learns in the proper manner; and him who has grasped all these things in this way, as he himself lays it down, he

calls most truly the wisest, and he strongly affirms it in play[12], and he enjoins upon those who are eager to philosophise that these mathematical sciences are to be pursued and chosen at all costs, whether they be hard or easy; and very reasonably so, if indeed apprehensions of the continuous and the discrete come about through these alone, and the cosmos and all the things in it are composed of the continuous and the discrete.

32

Now the exact apprehension of these is wisdom, the desire for wisdom is philosophy, and philosophy, alone among all arts and sciences, secures for man the end that is proper and natural to him and leads to the happiness which, in distinction from the other animals, belongs to him alone and which is by nature earnestly pursued as the end most fit to be his aim.

Chapter 8

What the common criterion of all the mathematical sciences is, and how it is discovered from the section of the line which the Pythagoreans hand down.

Next after this we must speak also of the criterion of all the mathematical sciences, what sort of thing it is and what differences of activities it has within itself. Taking the matter up from the beginning, then, let us conduct the whole of our instruction about it on the basis of division.

All intelligibles, then, are divided into two: into those called intelligibles in the proper sense and objects of science, and into the objects of discursive thought; and the intelligibles are first, while the objects of discursive thought are second and inferior. Again, after these there is another essence, that of sensibles, and of these some are sensibles in the proper sense, which are also objects of opinion, while the others are objects of conjecture. Objects of opinion and sensibles in the proper sense are the particular bodies, such as stones, pieces of wood, the four elements, and these are first among sensibles; after

them come others that are weak and not like them, but are consequent upon the first. These are shadows; for shadows are concomitants of bodies, and if they did not have some other body lying beneath them, they would not appear. Shadows, then, and the things in waters and in mirrors are images, existing in other things and not by themselves, nor appearing apart from other things, but having fallen upon other bodies, and when these are withdrawn they do not appear.

33

Hence they are sensibles in genus, because they fall under sense-perception, but they are objects of conjecture and of belief rather than things having real existence, being so called in accordance with the belief that attaches to things which are not demonstrative but are otherwise taken up for acceptance, out of belief in those who put them forward. For shadows do not have from themselves that which makes for apprehension, but from the bodies upon which they have fallen and in which, coming to rest, they appear. Such belief, then, is unstable; for if these things were to depart from the mirror or the water or the ground, they would be nothing at all. So that, whereas even bodies are objects of opinion and possess their being in seeming, shadows rank still lower, in that they do not have solidity from themselves but rest upon something else. Now the objects of discursive thought too resemble these, standing to the objects of science and the intelligibles in the same ratio as objects of conjecture to sensibles and objects of opinion. For the intellect holds the Ideas, which are the things that truly are, as it were by contact, whereas the objects of discursive thought, which are the geometrical objects, are beheld by discursive thought, discursive thought no longer drawing near to them directly and as it were by immediate apprehension, but its approach to them coming about rather through reason, and as though we were descending from the Ideas to likenesses of them and intelligible images.

34

And likewise the objects of conjecture, those in shadows, rank below sensibles, in that the latter fall under sense-perception by themselves and in a direct line, while the former are observed in another, upon another and on account of another. For the shadow certainly does not exist by itself, but either on the ground, which is sensible by itself, or in the mirror or in the waters, which were, as we saw, sensible by themselves. In this way, then, mathematical objects too seem to appear as it were in the Ideas, and to have their support upon them; for they must not be conceived by abstraction from sensibles, but as having descended from the Ideas and having their character as images from them, by having taken on magnitude in addition and by appearing in extension. For what faintness and want of any support in itself are in the images of sensibles, that bulk and extension are in the intelligible ones; but since this too hastens towards what is without bulk and without parts, it seems to come to rest in the partlessness of the Ideas, as shadows do in the resistance of sensibles. Just as the objects of discursive thought, then, are separated from the intelligibles, so too is discursive thought from intellection. That is why Brotinus too, in his On Intellect and Discursive Thought, separating them from one another, says the following: 'Discursive thought is greater than intellect, and the object of discursive thought than the intelligible; for intellect is that which is simple and incomposite and which primarily thinks and is thought (and such is the form; for it is without parts and incomposite and prior to the others), whereas discursive thought is that which is manifold and divisible and which thinks secondarily (for it has taken on science and reason in addition)[13], and similarly too the objects of discursive thought; and these are the objects of science and the demonstrables and the universals, which are grasped by intellect through reason.'

35

In these words he calls discursive thought and the object of discursive thought greater, not in power but in multitude (and these stand in inverse relation to one another), and he marks them off from intellect

and the intelligibles not by these alone, but also by the fact that the one set are simple and incomposite, the other multiform and composite; and because the one set think and are thought primarily, the other secondarily and as receiving from the former the activity of doing so; and the one set are established in forms, while the other exercise a manifold activity in reasonings; and the one are indivisible, the other divisible; and the one are superior to demonstrative syllogism, while the other reason syllogistically to some conclusion about beings; and the one are the beings themselves, while the other have embraced the particulars in the universals and display them along with these; and the one employ immaterial and pure activities, while the other have their intellection mixed; for they grasp the objects of knowledge proper to them by intellect through reason, or by intellect together with reason. It follows from this, then, that both the things judged and their criteria stand apart from one another, so that the objects of discursive thought differ from the intelligibles, and discursive thought from intellect.

35

Still more clearly does Archytas, in his On Intellect and Sense-Perception, distinguish the criteria of beings, and he sets forth the criterion most proper to mathematical objects in these words:

36

'For in ourselves', he says, 'there are, in respect of the soul, four kinds of cognition: intellect, science, opinion, sense-perception; of which two are principles of reason, namely intellect and sense-perception, and two are its ends, namely science and opinion; and like is always capable of knowing like. It is clear, then, that intellect in us is capable of knowing the intelligibles, science the objects of science, opinion the objects of opinion, and sense-perception the sensibles; for which reason, then, discursive thought must pass over from the sensibles to the objects of opinion, from the objects of opinion to the objects of science, and from these to the intelligibles; and these are to be brought into concord, and when contemplated through themselves

they are true[14]. With these things determined, we must consider what comes next. For just as a line is cut in two, and again equal segments[15] of each part are cut in the same ratio, so too let the intelligible be divided in relation to the visible, and again let each be marked off in the same way, and let them differ in clarity and obscurity relative to the others[16]. In the same way, then, of the sensible one segment is the images in waters and in mirrors, and the other part is the things of which these are likenesses, plants and animals; and of the intelligible the part that stands analogously, as the likenesses do, is the genera with which the mathematical sciences are concerned.

37

For those who are concerned with geometry, having hypothesised the odd and the even and figures and three species of angles, work out the rest from these, but leave the things themselves alone as though they knew them, and are unable to give an account of them either to themselves or to others; rather, they make use of sensibles, as is natural[17], yet it is not these that they are inquiring into, nor is it for the sake of these that they construct their arguments, but for the sake of the diagonal and of the square itself. The other segment of the intelligible is that with which dialectic is occupied; for dialectic itself treats the hypotheses as really hypotheses, but as starting-points and stepping-stones[18], in order that it may come as far as the unhypothetical, to the principle of everything, and, keeping hold, may descend again to the conclusion, making no additional use of anything sensible, but of forms themselves through themselves. And to these four segments it is well to distribute also the affections of the soul, and to call the one at the highest intellection, the one at the second discursive thought, the one at the third belief, and the one at the fourth conjecture.'

37

I think, then, that through these words too it has become quite plain that there are four differences among beings and four principles of judgement, and that reason, holding a middle position, lays hold of

the two extremes, the intelligibles and the sensibles, being set in the rank of an end in relation to intellect and sense-perception, since these are its principles, and being brought to completion by them.

38

This too is a common axiom concerning every cognitive faculty, that like things are known by like. It is possible, then, to learn both from both, and the one set of these from the other, and possible to make the equal divisions in their case both in common and in particular, and an order of transition from the one set to the other, that is, from the inferior to the higher; and he has determined how one must effect the ascent and co-ordination of all things towards intellect. Next after this he cuts the line, which is one, in order that we may conceive of the cognitive faculty as one, but he divides it in two in accordance with the primary differences among beings and the judgements, divided in two, that bear upon them. And he posits these divisions as equal in virtue of their participation in the reason-principles and the forms, and on account of the likeness of the things that participate to the things participated in, and because the proportion is in a way the same in both. Again, he divides each of the segments in the same ratio, since the cognitive faculty is throughout of the same form as itself, and he makes its differences lie in clarity and obscurity and in being perfectly or deficiently determined, and he exhibits their distinction from one another, showing in what respect the second deviate from and are inferior to the prior. First he divides the sensible, as being more familiar, and takes from it the existence that appears, by way of likeness, from the sensibles, namely the images in waters and in mirrors, cutting it off as a single nature.

39

The other part he marks off as the true one, the things of which these are likenesses, such as plants and animals. For it is from these that conjecture of the images arises, sense-perception being turned towards the latter by reflection and so knowing them secondarily, just as they also exist in a secondary manner, whereas it applies itself to

the sensibles immediately, just as they also exist primarily and have their enmattered existence in themselves. From the proportion of these, then, it is possible to understand the other segment as well. For the genera with which the mathematical sciences are concerned are analogous to the likenesses, and the cognitions of them have a certain likeness to the conjectures of the images; for they receive their activity from intellections and pass over from the intelligibles to mathematical objects as to likenesses, and they make use of hypotheses and do not know the cause. And this is the criterion of mathematical objects, being a faculty of prior knowledge[19] of a different thing, and not of the intelligible, and apprehending the object of discursive thought by a different cognition, and not by intellection; for intellection is the criterion of the dialectician, and through it he contemplates beings and the forms and all things unhypothetical, and is able to give an account of all things, and makes no additional use of anything sensible, but of the intelligible forms. Since, then, the judging faculties are four, a certain order among them is observed and their activities are in fact divided: at the highest intellection, at the second discursive thought, at the third belief, and conjecture at the fourth.

40

From these considerations, then, on the basis of division, it has been sufficiently shown what the criterion of the mathematical sciences is.

Chapter 9

On those who assign a determinate essence to the mathematical sciences, of whom the first opinion set out is that of those who refer it to soul; several reasons for such a hypothesis are stated, and starting-points for the whole theory concerning them.

But if we must now go on to grasp in a determinate way as well the form of mathematics, both what it is and how it subsists, let us look first at the opinion of those who refer it to soul; for upon this one could fix one's thought in a determinate way.

Now one would not reasonably posit the soul as one single genus among the things that exist in the mathematical sciences, on such an approach to the theory; for in that way the knowledge of mathematical essence would become divided. Hence one should not define it separately as the form of what is extended in every direction, or as a self-moved number, or as a harmony subsisting in ratios, or as anything else of that kind; rather it is right to weave them all together in common, on the ground that the soul is both a numerical form and subsists in accordance with numbers that embrace a harmony; and all the commensurabilities in common, as many as fall under mathematics, are to be ranked under it, and the proportions as a whole are to be placed under it. For this reason, then, it coexists with geometrical, arithmetical and harmonic proportion all together, whence indeed it is also identical with the ratios that accord with proportion, has a certain kinship with the principles of beings, lays hold of all beings, and is able to be made like to all things.

41

Such, then, are the reasons for this sort of supposition. But starting-points for the theory, the mathematical and that concerning the soul together, would arise if we were to observe that everything limited and determinate reaches it from numbers, and the unitary reason-principle from the nature of the One, while the power that proceeds to magnitude and increase and has abundance, so as to give itself to all things of due measure, is present to it from geometrical essence; and the power of harmonious motion, and order, and the commensurability of things irrational[20], and the good measure found in numbers that are concordant or embrace concord, come to it from essential harmony. That is why the soul both hearkens to harmonies and delights in things attuned, as being itself too a harmony, and has its essence from numbers and other such mathematical measures, which have received a kinship both with the intelligible forms and with sensible essences and the enmattered forms; for the present opinion gives a starting-point for theory in relation to all of these,

since the mathematical opinion thus hypothesised is sufficient to furnish all such conceptions. To sum up the whole opinion: we conceive the soul as existing in reason-principles common to all the mathematical sciences, possessing on the one hand the capacity to judge them, and possessing on the other the capacity to generate and produce the incorporeal measures themselves, to which one can also fit the generative production of the enmattered forms and the fashioning by means of images, which proceeds from the unseen into the manifest and joins what is outside to what is within.

42

For in all these respects, to put it comprehensively, the reason-principle of the soul contains of itself the entire complement of the mathematical sciences.

Chapter 10

How the essence of the soul is constituted out of all the mathematical sciences, and by what determination their blending in it might be defined, and whether it contains in itself the whole existence of the mathematical sciences or some other principle of them is also observed.

But whether it is a mixture of all the things that exist in the mathematical sciences, or itself brings them all into existence in virtue of a single antecedent reason-principle, must be examined. If, then, it is a commixture from them all, the things that make it up, from whose coming together it is blended, exist before it, and it would no longer itself be the principle of mathematical essence; rather, since it is generated from mathematical sciences that subsist in scattered fashion and come together into the same thing, this, besides the other absurdities, also brings in along with the soul a kind of composition and an existence that is later-born, as arising subsequently out of certain prior things. If, however, it is itself primarily the originator of mathematical essence and brings this forth out of itself, it will be senior to it, and it both precedes it as holding the rank of cause and is superior as being other than it. But this too is contrary to the present

opinion; for in that way the soul would turn out to be more honourable than the things that exist in the mathematical sciences. It is better, then, to say that it neither precedes nor follows upon the things that exist mathematically, but runs together with them simultaneously and subsists along with them, having its commixture in them all and from them all as something incomposite and indivisible, being present to them in a uniform manner and participating in them all in a unitary manner, having gathered into itself a power that embraces them all and having given itself in the same way to all the mathematical sciences.

43

And if this is so, it has also embraced all things in itself perfectly and without deficiency, and leaves nothing outside itself (for it is itself perfect, and it is not possible for any of the things that are to be left without its own principle), and in this way the essence will be one, since the same principle extends through them all. Differences, however, there will be none the less, in accordance with the different powers, lives and activities of the soul and the multitude of its essences, which is contained in one. Such is the determination that one might reasonably lay down concerning these matters too.

43

Let so much, then, be said by us concerning the essence of mathematical theory.

Chapter 11

What the work of mathematical theory is and how it comes about, and that it is named 'mathematical' in agreement with these things.

The work of this science is not determinate, nor is it in the same condition and in the same way, like that of intellect, nor does it altogether have its knowing from itself, as that sort of thing is connate with intellect; rather it is roused from outside towards its cognitions, and, receiving from others the starting-point of recollection, it thus

projects it from itself; nor is it stable in a single activity, like that of intellect, but rather proceeds in motion from itself and to itself. Nor yet is it full of itself, like what is intellective, but in seeking and finding it always advances from a certain emptiness of knowing to the filling of itself. And it is likewise held in the middle between limit and unlimitedness; whence it always advances from the unlimited towards being determined, and passes over to partaking of the mathematical forms.

44

It is for all these reasons, then, that this science comes about with learning first leading the way, the starting-point of which is supplied by the teacher, and discovery then following, which is attached to the starting-points laid down by the teacher; for it is in accordance with these that the soul recollects the true forms in mathematics and projects the reason-principles proper to them. Sometimes, however, the activity also comes to be a common one, arising from both, which is why Archytas in his On Mathematics says: 'For you must come to know the things of which you were ignorant either by learning from another or by discovering them yourself. Now learning is from another and by another's means, while discovering is through oneself and one's own; and to discover without seeking is difficult and rare, but with seeking it is accessible and easy, whereas for one who does not know, to seek is impossible.' For in these words he put learning first, as the starting-point of this kind of science, and pointed out what is proper to it, namely that it is imparted by another. Second he added 'discovering them yourself'; for even if this takes precedence in power, still in the human order at least, relative to us, it is second; for those who have fallen into generation must first be reminded by others. Now it is possible to take these as two ways of receiving science, but it is also possible to comprehend them in reasoning as one; for whenever we have acquired the mathematical sciences as from another and as belonging to another, we then bring them to hand ourselves from ourselves as our own.

45

This is easy to learn from discoveries; for it is as having them in ourselves that we discover them, and recognise them once they are discovered. The same thing is evident from acts of seeking as well. For if it is impossible for one who does not know to seek, there was a time when we knew these things, and it is not this present time (for now we are ignorant of them); therefore we knew them previously. And for this reason the mathematical sciences are accessible and easy to discover for one who seeks, but difficult and rare for one who does not seek, because they are in a way present in souls, and at some earlier time were with them in scientific knowledge that was in activity. The study that proceeds by way of the mathematical sciences, then, is a road from seeking to discovery, and from learning to seeking and discovery. Hence indeed it has also got this name, that of being called mathematical (mathēmatikē). For from that from which the science first receives its starting-point and without which it cannot arise in us, I mean learning (manthanein), from this it has taken its name.

45

Let these points too about it, then, stand as thus clarified by us.

Chapter 12

What the powers of mathematical science are, what orders they have among themselves, according to how many differences they are divided, and in how many ways they are conceived.

As for its powers, one might enumerate many others besides, but among the first let there be considered those that gather from multitude towards the cause that is the same and unified, and all those that divide from the one into multitude.

46

For since they are constituted out of limit and unlimitedness and out of the One and multitude, and are intermediate between the divisible and the indivisible, they have also, in accordance with the same account of their essence, a share in both gathering and division, in

both unfolding and rolling together, and in both reversion to the determinate and departure from it. Now this sort of activity of contemplation is found in dialectic too, which is concerned primarily with being; nevertheless this science does differ from that one in no small respect: for the one contemplates being without qualification, and gathers or divides this, whereas the other examines mathematical being and exercises these twofold activities of reason upon it. There are also other powers which look to what is common over many things, all those that contemplate in the differing mathematical sciences certain common forms and common ratios and common measures by which the things that differ are marked off, such as the studies of equality and inequality and those of the commensurable and incommensurable; for these powers contemplate what subsists in common over several things. Opposed to these are those that contemplate what is proper to the essence of each thing, for instance of numbers in so far as they are numbers, and of magnitudes in so far as they subsist as magnitudes, and in the same way in the other cases. Among such powers are also those that apprehend the proportion that subsists everywhere, beginning from above with the very first things, ending at the last, proceeding through the intermediates, and everywhere preserving the same ratios, and different ones in things that differ, and rendering them manifest throughout the whole. Moreover, those that contemplate the beauty and the measure of mathematical essences, and what is attuned and commensurate in them, must also be held of some account; for they have by their own proper nature good order and perfection and all the goods that befit mathematical forms.

47

Now to most people these powers seem both to be unmoved themselves and to be active about objects of knowledge that are unmoved; but this view is not correct. For there are certain mathematical sciences which examine the number of motion and its measures, how they stand both in themselves and in relation to one

another in point of order and commensurability, and the incorporeal revolutions of the soul, with which the circuits of the heaven also coexist, how they stand in point of commensurability and according to what numbers, and why they fit together, and which examine all such matters; among these, indeed, astronomy and harmonics are included. In order, then, that the mathematical powers may extend also to those that contemplate motions, these too must be posited as contained under mathematical science. Over against these one must distinguish the stable powers, which contemplate the unmoved forms and ratios and are able to reckon out[21] the order of these relative to one another; among them are the majority of the powers connate with mathematics. Their orders must be marked off according to the essence of the objects of knowledge which they contemplate, whether some take precedence and others are subordinate, and according to the ranking in honour of the beautiful, whether some contemplate the most senior and highest beauty and others that which is inferior and imperfect. The differences among them are to be taken from the manner of their activities and the variation in their knowledge, and from the different beings that are yoked with them, with which they interweave their acts of learning.

48

The question in how many ways they are conceived must be examined from the varied objects of knowledge that learning has; for from these the powers are seen both to subsist themselves in many ways and to exercise their activities in manifold fashion.

48

In this way one might mark these things out at the beginning, as in an outline. But, as is fitting, one must await the full account of them; for in that way above all the teaching about them would be handed down complete.

Chapter 13

What the elements and genera of mathematical science are, and in what way the same things are elements and in what way genera, and in what respect these differ from those in the other sciences and essences, both the intelligible ones and all that are in generation.

Since every theory and every science derives its unshakeable character from its first elements, when these are determinate and never otherwise, and attains its most perfect comprehension from the passage through the elements, and since, further, each finds from its own proper genera the manner of arguments and demonstrations appropriate to itself, it is necessary in the case of mathematics too both to distinguish beforehand those elements of it that are most common to the whole concerted effort[22] of the mathematical sciences, and to hunt for the genera that are most proper to it and that most of all extend in common to everything. And when we have discerned these, let us consider in turn whether the elements will be one thing and the genera another, or whether the same things are viewed in one way as genera and in another as elements, and in what respect such things in mathematics differ from the genera and elements in the other sciences and essences, both all that are intelligible and all that are borne along in generation.

That the elements and genera of mathematics are determinate and always at rest is both agreed by the best of those engaged in philosophy and clearly attested by the mathematical demonstrations themselves, which are always in the same condition and in the same way.

49

And that the account of these also agrees with the first principles of mathematical essence is easy to learn; for here too the One and multitude, limit and the unlimited, sameness and otherness, are elements and genera of the science and of the things known by it. Whenever, then, we consider these as causes and as productive of the

whole of mathematical essence and of the theory concerning it, let the causes just mentioned be conceived as principles; whenever they are conceived as inherent and as completing the essence and the account of the science, let them be conceived as elements; and whenever we discern them as common, holding of all the mathematical sciences, furnishing the bond of essence for the particular ones and none the less subsisting themselves by themselves, then let us view them as genera. The same things, therefore, are in one way originative of mathematical theory and of the things known by it as beings, in another way are conceived as elements, and in yet another way again as genera; not because they differ conceptually, in account alone, nor because they change along with their different relations and turn from one thing into another, but because the same things make several processions and differences within themselves. And indeed, what is without difference[23] is also disposed in accordance with the difference of cause; for it is not the same thing for one of the incorporeals to be by itself and to be completive of other things, nor again have that which is productive of certain things and that which contributes in any way whatever to essence obtained the same rank.

50

Nor is there any way in which that which is co-ordinated with other things ever ranks the essence it has by itself in that co-ordination; rather, it leaves that essence apart, and weaves the other into the completion of those essences to whose being it contributes. By the same reasoning, then, the cause that furnishes being does not coexist with the things brought to completion by it either, but is senior to them in the very account of its essence, and has in itself a separate existence, on account of which it also gives to the things torn asunder by it[24] another existence, after itself. In this way, then, the unlimited and limit have been reasonably marked off by us among principles and elements and genera alike. These differ from the intelligible principles, elements and genera in that they fall short of them in

perfection and purity and simplicity, and in the comprehensiveness that extends furthest, in being determinate, and further in beauty and in all goods; while they surpass those in generation in order, in commensurability, in their unmoved and stable nature, in their pure participation in forms, in their incorporeal and immaterial nature, and, to speak comprehensively, in all the better attributes. From this, then, it is inferred that they are intermediate between both of these, and hold a middle rank that is able to have communion with both and to ferry across to both of them equally.

50

One who came to such determinations about these matters too would not miss what is fitting.

Chapter 14

On mathematical likeness and unlikeness: what they are, how far they extend, and how they are present in mathematical essence; and in what respect they differ from the genera of the same name that are spoken of among intelligibles and among sensibles.

As for likeness and unlikeness, everyone would agree that there is a great deal of it in the mathematical sciences and in mathematical essence, and that it has great power there; for it is not even possible to know any theorem mathematically unless one constructs it by defining some figure like it, conducting the argument about it by means of another image, and constructing one thing from another according to a single account, that of likeness.

51

But what deserves study is this: what these common features of the like and the unlike are, how far they extend in the mathematical sciences, how they are present in them, and in what respect they differ from the things homonymously called like and unlike among intelligibles or sensibles. Now this must be understood: that in the case of mathematical essence the like and the unlike are not spoken of

in respect of quality, nor in respect of figure of the kind that supervenes in certain cases as one thing upon others; for qualities of that sort tend to occur in composite things and in connection with composition, where the substrate is one thing and the accident in the substrate another, working a character and a specific form into the underlying nature; whereas the like and unlike that we are now investigating is senior to all composition. Nor again is it viewed in terms of a relation of the kind that subsists in being disposed in a certain way; for one must not conceive the existences of things that exist by themselves as depending on other things. Let it be laid down, then, that the like and unlike now under discussion holds primarily in respect of essence, not every essence, however, but mathematical essence. These, then, will be certain forms of mathematical essence. For one ought not to set quality and quantity in a division over against essence, and then assign one theoretical science to essence and another to quantity, and mark off this latter as being mathematical science; rather, just as its nature is, so too one should examine the proper mathematical essence, and these forms, how many they are and of what kind; and the quantity which the science does also examine along with it is neither the quantity in bodies nor the intelligible paradigm, but so much as is mathematical; in the same way, then, it is naturally fitted to consider along with it the like and the unlike too, whether as common genera or species concerning the essence, or as common powers generative of the forms in the particular mathematical sciences, according to the very account of being.

52

They extend, in the case of mathematical essence as a whole, to the whole of its being, and in the case of the particular mathematical sciences, to their divided existences; and thus, to whatever extent one considers more or fewer, greater or lesser circumscriptions of these, the like and the unlike are coextensive with them all, reaching either further or less far. For it is altogether impossible for multitude or

division or unification or sameness and otherness to exist among genuinely mathematical objects unless essential likeness and unlikeness have gone before. Let us no longer wonder, then, that they extend to one genus and to more and to fewer and to all; far rather ought we to consider this, that they are present in each in a manner akin to it, according to the proper character of each. And this too must be considered, I mean the one and the many of the like and the unlike and what lies between these, and one must look to what their fitting order is, and to their distribution among each of the mathematical sciences, according as each of these is naturally fitted to partake of them.

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If this is so, then that mathematical science would be the most authoritative of all which discovers the same cause of likeness and of unlikeness alike in prior and in posterior things, and, in the case of things of the same rank, makes the due distinction between them. Moreover, contrary to the opinion of the many, and after the manner of incorporeals, it contemplates the like in things that differ and the unlike in things that do not differ. And in the same way it examines the contraries together in one another, observing, for instance, the unlike in the like. How, then, each of them comes to be present we should best understand if here too we were to preserve the peculiar character of the essence that is the subject of our account. For let us not conceive them as coming to be present in the way that enmattered forms appear as images about matter (for they are connate with mathematical essence, in which they have their being, and cannot be removed from it), nor yet in the way of things innate in bodies, such as heat in fire; for even though these latter subsist together as closely as may be with the things that have them, nevertheless a certain differentiation belonging to a composite is observed, inasmuch as the heat is participated in as one thing, while the things that have it participate as something other. But in the case of the things that pre-exist essentially in mathematical beings, a

simple essence is observed, incomposite through the whole of itself; for the more it is incorporeal and separate from composite and extended masses, the more does it possess the like and the unlike unmixed, in a simpler form, subsisting by itself and in greater purity. In the same way what is common in them too is unalloyed and uncorrupted. Since, then, we are agreed on these points, we must also grasp this: that mathematical likeness and unlikeness ought to be taken as different from the likeness and unlikeness found among intelligibles and from that found among sensibles.

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It will be distinguished from them in one way, that by which we distinguished the three essences (for it is clear that, since these differ in three ways, the forms in them too will differ in three ways), and in another way, because it is delimited by its intermediate nature between them, those being extremes and marking off the beginning and the end of the forms that concern their own essence; and in yet another way, let some be conceived as originative, others as arising in the rank of an effect, and others as growing up between the causes that lead and the things brought to completion as last. Having grasped these points in advance, we should easily be able, starting from them, to study in due order also the like and unlike things proper to each of the mathematical sciences, whenever we give the account proper to them; for the present, let what is common concerning them be taken as stated by us thus far.

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Constituted, then, from genera, principles and elements of such a kind and so many, and being such as we have already described it, mathematical science does not extend only a short way, nor to a mere few of the affairs of life, but contributes to the greatest and finest of goods, both divine and human.

Chapter 15

How the whole of mathematical science, both itself and its genera, its elements and its principles, extends into the whole of philosophy and the parts of philosophy; and how it has communion with them, and in virtue of what contribution.

First, then, let us try to state that it extends over the whole of philosophy and over all its study of both the things that are and the things that come to be, both it itself as a whole and its genera, its elements and its principles, and that all the genera or species of mathematics that there are have permeated the whole of philosophy.

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Hence indeed these men make use of the mathematical sciences everywhere, whenever they pursue any philosophical study. For being incorporeal and intermediate, and capable of being fitted to all the things that are and of being assimilated to them, the mathematical sciences give us great help towards all the sciences within philosophy. For theology they make ready in advance a preparation and a fitness, and they furnish a likeness to it, a leading up and a thorough purification: one that releases the intellective organs from their bonds, purifies them thoroughly and joins them to being; that by the beauty and good order of the things contemplated in the mathematical sciences draws near in some way to the intelligibles; that through the contemplation of things unchanging and unmoved is assimilated to the intelligible and determinate things that stand at rest in the same respects and in the same way; that accustoms discursive thought gently to apply itself to the brightness of being, leads it away from bodies, implants belief concerning the essence of incorporeals, and provides scientific firmness and exactness. For all such things afford a great starting-point for the understanding of the things that are and are intelligible. Moreover, mathematical science assists the natural philosopher in no small measure, exhibiting the commensurability of the things in nature, their surpassing good order,

and the proportion that extends through all the things in nature, and examining beauty and natural forms and the accounts concerning them, elements and the simplest things and their figures, and the most important genera and species; for all these are employed by those who genuinely conduct the study of nature from the first principles.

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To the statesman, again, it contributes by being the guide of an ordered motion of actions, by supplying a motion of the theorems that stand at rest, and by implanting in all equality and the fitting agreement. And it assists the moral philosopher by containing the reason-principles of the virtues and by displaying mathematical forms as paradigms, of friendship, for example, or of happiness, or of some other of the greatest goods. It also sets forth mathematical paradigms of all the things in life, such as fertility and sterility, fruitfulness and barrenness, and everything of that kind. Hence indeed one must use the mathematical sciences everywhere, sketching out philosophy in them as in paradigms. We do not, however, take the same paradigms everywhere, but in each case adduce the appropriate ones according to the proper genera of the science. So then the whole essence of mathematics, both itself and the genera and elements in it and all the principles there are, extends over the whole of philosophy. For it is possible to extend mathematical reasonings in common over the whole of it, and it is possible also to extend them to the parts of philosophy, in whatever way the need of the argument may require. It has communion with them in so far as it has a certain likeness to them, and the contribution it makes to them is one that conveys us across to them and guides the way. To the forms that are at rest and determinate, that do not at one time exist and at another not, but are always in the same state, it is of a nature to refer and to gather us, inasmuch as it falls short of them in perfection and purity and in the fineness, so to speak, of incorporeality, and is assimilated to them as to things that surpass it;

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while of the enmattered forms in the realm of generation it holds out paradigms separate from bodies, in mathematical forms; and thus it co-operates in both directions.

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Such, then, is the contribution it provides to philosophy as a whole and to its parts.

Chapter 16

How many goods it contributes to the arts, both to all of them universally and to those distinguished by genera, such as the theoretical, the productive and the practical; and instruction about them in summary.

As for the arts, into all of them, to speak simply, it implants a scientific discernment, exhibiting their principles, ends and determinations, teaching thoroughly their measures and the judgements of them, distinguishing what is correct in them from what is mistaken and marking off the appropriate elements of each of these, knowing their end, imparting exactness to them and bringing about their discovery. For since this science contemplates essence apart from matter and employs reason-principles that are separate and not disturbed by enmattered things, it is with good reason, on that account, more of a cause and more of a guide than the arts that are in contact with matter, with respect to their discovery, their judgement and their discernment. The theoretical arts, then, it thoroughly purifies and perfects; to the productive arts it is attached in the rank of a paradigm; the practical arts it rouses and moves by its own forms, which are at rest; and in all of them in common it fits the separate reason-principles to the enmattered forms. And being, as it were, the master-art of them all, it accordingly takes the lead over them;

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it is the art that makes use of them, and it renders them honourable and beneficial once they have been made secure by mathematical

reasoning, and by mathematical demonstration strengthens their reasonings and makes them free from falsehood.

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Mathematical theory has thus been shown to extend also over all technical practice and knowledge.

Chapter 17

What the order of training in mathematics is; whether it has an order by nature and one relative to learning; and whether each order agrees with the other and the two with one another.

Furthermore, that there is in it a twofold order, the one coexisting with it by nature, the other relative to learning, is easy to grasp from the following. For if orderliness, and the fact that this follows of necessity upon that, comes to all other things from mathematical science, then surely far more does mathematical theory itself contain order within itself and conduct in orderly fashion its course of training towards what is perfect. Now the natural order of the mathematical sciences places the simpler first as prior, arithmetic, for instance, before geometry, and sometimes the same things take precedence in teaching too, when the learning of composite things proceeds from their elements; nevertheless, sometimes, relative to us, composite things will be prior to simpler ones for the purpose of learning, when they are better known: for example, the heaven as a whole and the motion belonging to it is surely better known than the sphere taken simply and the sphere considered just as moving. If, then, someone should show what is unapparent by means of what is apparent, such a mode of approach will not be one to reject. These being thus divided in two, both modes should be used, the one as more scientific, the other as better known.

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And when it is necessary to use only one of the two modes, one must prefer the one that is more appropriate and contributes more to the

object of scientific knowledge in question; but when it is possible to use both, one ought to guide the way to scientific knowledge through both. Hence in many mathematical studies the same problems are demonstrated both by analysis and by synthesis. In cases, then, where the two modes of scientific knowledge agree, both should be used. One must also aim at the state of each person: whether, for example, someone who is naturally gifted and quick is able to pass easily from one thing to many and to take in all at once many things that have some kinship with one another. This too must be considered: what exactly is the end to which the occupation with mathematics is referred, whether it is simply the learning of the theorems of the science, or whether one is leading them up to philosophy and proposes to be guided through them to the vision of the intelligible; for such a person the order would be different, sometimes overstepping the natural sequence of the mathematical sciences. Again, each of the objects of contemplation in mathematics presents as prior what is evident at once and less perfect, for example, that the right-angled triangle has its hypotenuse equal in square to the sides containing the right angle, whereas the more perfect points, which require extraordinary demonstration, are handed down later: all that, in connection with the right-angled triangle, bears on the motion of the stars, on their relation to the zodiac, and on the motion of the sun and moon.

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So too with what concerns harmony: the matters concerning simple harmony are taught first, those concerning the harmony of the cosmos later.

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We have said these things in advance for this reason: that, employing a definite method in the order of the treatment of mathematics, we may aim at two things, the nature of the things and the capacity of the learners, and may use each fittingly, and, whenever these agree with one another, both equally.

Chapter 18

What the proper modes of the Pythagorean transmission of the mathematical sciences are, how they used them and with regard to whom, and that they always applied what was appropriate both to the things and to the learners.

Moreover, the proper modes of the Pythagorean transmission of the mathematical sciences possessed a wonderful exactness, and differed greatly from the technical instruction given by those who spend their time on the mathematical sciences. Let us therefore sketch that transmission in outline, as best it is possible to speak of it in a common account.

Let this one point, then, be agreed: that it was by starting from above, from the first principles, that they made their first construction of the mathematical theorems, inasmuch as they made the undertakings of discursive thought proceed from the very first essence of those theorems, and led the whole mathematical apprehension back up to that essence as its final term. Further, and following upon this, they made it their practice to exhibit first the discoveries of the theorems, and to use nothing as though it were already given, but in every case to consider how that which is proved in the mathematical sciences might come into existence. There was also another mode among them, the mathematical mode through symbols, as for instance the pentad is the symbol of justice, because it signifies symbolically all the species of what is just.

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This form was useful to them for the whole of philosophy, since they taught most things symbolically, and held this mode to be proper to the gods and suited to nature. But that they did also hand down the first principles and the discoveries of the mathematical sciences is clear from the other mathematical sciences as well, and is manifest also from the arithmetical methods. For they teach thoroughly how each genus and species of numbers is primarily generated and how it is discovered by us, on the ground that the study of these things is not

scientific unless one grasps them by starting from above. Further, to the things that truly are and to all things divine, to the states and powers of the soul, to the phenomena in the heaven and the revolutions of the stars, to all the things in the realm of generation, both the elements of bodies and the things compounded out of these, and to matter and the things generated from it, they always brought the mathematical theorems into kinship, both all of them taken simply, and by taking from each one the imitations appropriate to each of the things that are. And they made the references of the mathematical sciences to the things that are either by a sharing in the same ratios, or by some faint reflection, or by a likeness, whether closely approaching or standing far off, or by a certain representation in images, or by an antecedent cause, as in a paradigm, or in some other way. And in other ways too, of many kinds, they yoke the mathematical sciences to things, on the ground that things are able to be made like the mathematical sciences, and the mathematical sciences are of a nature to be likened to things, and both are reciprocally assimilated to one another.

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In variety of argument, then, and in abundance of methods they took no very great delight, regarding this as too much a matter of mere reasoning and as standing apart from the truth of things; what they primarily welcomed was the knowledge of the problems themselves, as contributing to the science and the discovery of the things that are. And they relied rather on the discovery of truth and on the apprehension directed towards things, and not on the keenness and sharpness of the syllogisms concerning the problems. Hence they did not pride themselves on an abundance of mathematical arguments either, but were seen to give preference to that which contributes to the discovery of things.

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These, then, and others of this kind were the modes of mathematical transmission among them. And they used them scientifically and in

company with theoretical philosophy, aiming at the things that are and at the beautiful, thinking that they ought always to give precedence and honour to what is limited and to what is gathered into the briefest compass, and selecting from them whatever was useful with regard to themselves and their associates and to the whole science of the things that are. Further, in handing them down they aimed, in one way, at the things, according to the order these possessed and their continuity with one another (for it was in accordance with a sequence of this kind that they marked off the first and the second theorem among them), while in another way they looked also to the learners, and aimed at these: what their capacity is and how they will be benefited by them, and what is to be handed down to beginners and what to those who are making progress, and which mathematical teachings are esoteric and which exoteric, and which may be spoken and which may not, and to whom they are to be handed down together with scientific knowledge of the things, and to whom as just this alone, mathematically.

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For exactness in all these matters was not idly practised among them, but in order that the treatment of mathematics might hold fast to one thing, the beautiful and good, and be ordered towards one thing, the science of being and assimilation to the good. And indeed in this way it was not merely a bare knowledge of the mathematical sciences that was handed down, but a life befitting them was ordered together with it, and an ascent to the most honourable things was duly established through them. For this very reason it is worth while to practise in this way the Pythagorean occupation with the mathematical sciences, as being exceptional and preferred before all the mathematical arts.

Chapter 19

A division, according to the Pythagoreans, of the whole of mathematical science into its most important genera and species, making the theory of them a common one.

But since one must not only examine its good as a whole, but must also take up its genera and species, how many they are and of what sort, let us make our instruction about them a common one, capable of extending in the same way both to them all and to each of the mathematical sciences.

Of all that is mathematical, then, and of what is proper in each case, of whatever sort it may be, the first object of contemplation is the theological, which is fitted to the essence and power, the order and the activities of the gods by a certain suitable representation, and which is in fact held most worthy of serious attention among these men: for instance, in the case of numbers, what sort of numbers are akin and of like nature to what sort of gods; and in the case of the other mathematical sciences too it is their custom to think in the same way. After this, the mathematical sciences among them endeavour to be active about the intellective being that truly is, an intellective circle and eidetic number; and many other such objects of mathematical study are contemplated in accord with the purest essence.

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Next, they bring the treatment of the mathematical sciences to a head in relation to the self-moved essence and the eternal reason-principles, defining that same self-moved number and discovering certain measures of the reason-principles according to certain mathematical commensurabilities. Much mathematical science is also to be observed in relation to the heaven and all the revolutions in the heaven, both the fixed ones and those of the planets, a science that examines together not only the varied motions of the spheres but also their uniform ones. And further, it treats of the enmattered reason-principles and the enmattered forms, both how they subsist and how they were originally brought forth; for of such a kind is that part of mathematics which separates in thought the shape and the figures from bodies. And in another way too it endeavours to give a physical

account of the things in the realm of generation, contemplating the simple elements and the reason-principles that attend upon bodies.

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All these parts of the method, then, the Pythagorean training uses in each particular and for all the mathematical sciences, and through them it brings about order and a thorough purification. For just as among mathematical objects the secondary are known from the prior, so in the case of the powers of the soul an ascent to the more perfect lives and activities comes about through them. Moreover, they neither neglect anything nor pass over any of the intermediates that go to complete such a science, nor yet do they leave the extremes unexplored. They go through them all without deficiency, and thus this science hands down, in the case of the most important and very first genera, the division which the science of division has shown forth.

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And from this it is possible also to discover the particular sections of the mathematical sciences, which we shall mention as we go on, in the account that is proper to it.

Chapter 20

What the definitional method of mathematics is and how it comes about, what benefit it contributes towards science, and that mathematics has an end, and what sort of thing its end is.

To this power of mathematics, then, there is another that answers as its counterpart, the definitional; for mathematics uses definitions too, and frames them with exactness. And the mode of the whole construction of definitions is this. When the divisive power of mathematics has divided according to genera and species the things in the mathematical sciences, then the definitional power gathers into one the differences that come from the division, and collects a single common account out of them all. It does the same thing also on the

basis of analysis; for when analysis has led intellection up to the simpler and more common things, and has distinguished the genera and the differences as each is by nature, then synthesis, which draws together, by bringing into one the things that differ and the simple things, defines each of the things in the mathematical sciences. And thus mathematical science has the definitional account as its own, from itself, and is able to discover it theoretically. Since, then, division makes the one many and the definitional power makes the many one, in both respects it is necessary that mathematics contemplate the one, from which it sets out and to which it is led up. This, then, will be the end, which one might also call form: a one over all the many theorems; and this same thing is also beautiful and good, and it is to this that the things divided hasten to be joined.

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For this reason, in the mathematical sciences too we always let go the unlimited and the particular, and hasten to ascend to what is common and determinate, all of us who practise mathematics in accordance with the doctrines of the Pythagoreans, until we have led the whole contemplation belonging to the treatment of mathematics up to the one that is the one of all the mathematical sciences together. And this is the end towards which those must hasten who are truly lovers of the spectacle of the mathematical forms as wholes.

Chapter 21

Who the founders were that led the way for mathematics according to Pythagoras, and what the special features of such a science are according to him, and how, following him, one ought to make one's arrangements concerning the mathematical sciences: a common discussion.

Since it is primarily Pythagorean mathematics to which we lay claim, and it is not possible to set this out completely in our account unless one discerns its first principles, it is necessary for this reason to include in our examination of the present subject those also who became for Pythagoras the originators of such study; for in this way

the inquiry into these matters would be most complete, having been made secure from above, from the first causes. They say, then, that Thales, who was the first to discover not a few things in geometry, handed them on to Pythagoras; so that whatever mathematical investigations of Thales we have received, these too we might justly appropriate to Pythagorean mathematics. After Thales, he associated for a long time with the Egyptians, and from them he obtained not a few good things for mathematical science; hence we should not be acting amiss in taking over along with the rest many of the things found among the Egyptians as well. And since later he associated also with the Assyrians and with those among them who are called Chaldaeans (for that is what the mathematicians are called among them),

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it is necessary that we take many things from these too for the mathematical method.

Chapter 22

What the peculiar practice of mathematical science according to Pythagoras was, and how many things useful to the soul and to human beings it had in view, and how they handled it throughout the whole of their own life.

Yet this is not sufficient. But since Pythagoras, having received the mathematical sciences from barbarians, added many things of his own, we must bring in such principles as well, and add the distinctive property of his mathematics. For he studied many parts of the mathematical sciences philosophically, and made them his own by his own apprehensions, although they had been handed down by others; he fitted to them the order that befits them, and made the appropriate inquiries concerning them; and he exhibits throughout a consistency that is always the same, so as nowhere to transgress what is consequent. Abiding by these principles, then, we must track out Pythagorean mathematics. And let us take as its special features, common elements as it were, the use of mathematical terms which, at

least on immediate hearing, is symbolic and strange; for since he was aiming at the things that are and at what is true, he accordingly also assigned to the mathematical sciences the names that accord with nature. And from these he made a beginning of instruction capable of guiding the hearers, if one should hear the names adequately, through adequate experience. Moreover, in the purity, subtlety and exactness of its demonstrations it surpasses all the study of the same kind pursued by others, and it employs great clarity and starts from what is well known; and the finest thing in it is in fact its loftiness of mind[25], led up as it is to the first causes, undertaking its learning for the sake of the things themselves and laying hold purely of the things that are, and in some places also joining the mathematical theorems to the theological ones.

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For so many are the elements that one might for the present put forward as common special features of such a science.

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As to how one ought to pursue the hunt for it, this is worth stating in its entirety, in conformity with what has been handed down by the men themselves. But since most things among them were matters of active practice and were preserved in unwritten memories, which no longer survive today, and about which it is not easy to infer or to discover anything, whether from writings or by hearing from another, we must do something of the following kind: starting from small sparks, we must always give such things body and make them grow, lead them back to their appropriate principles and fill up what has been left out, and aim so far as possible at the mind of those men, at what they would have said if it were possible for any of them to teach. And further, from the sequence of what has been handed down to us beyond dispute we are able to discover in a fitting manner the mathematics that comes next in order. For such modes of investigation will make us either attain the genuine Pythagorean mathematical science, or come as near to it as is in the highest degree

possible. And I have come to hold that the pursuit of this science, as it was practised in the manner of its own founder, is in agreement with it; for it was in every way peculiar, and exceptional beside the other disciplines, looking to the soul and to the purification of the eye of the soul, making discovery of the first forms and causes of mathematical essence, fitting it to the nature of the things that are themselves and assimilating it to the intelligible forms, and teaching thoroughly their kinship with the good and the affinity of the mathematical sciences with one another.

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Being of such a kind, then, the mathematical discipline sought out intensely, vehemently and without intermission the theorems that fall under it. And it contributed, for the soul, to purity of knowledge and subtlety of thoughts, to exactness of reasoning and contact with the incorporeal essences of its own order[26], and to due measure, good attunement and the turning round towards being; while to the human being it provides order for life, calm of the passions, beauty in character, and discoveries of the other things that are profitable for human life. And they handled it throughout the whole of their own life, weaving the benefit that comes from it into their actions and into the dispositions of the soul, into the organisation of cities and the management of households, into the works of the arts and into preparations for war or for peace, and in general they applied mathematics to all the parts of life, in a manner appropriate to the things, profitable to those who made use of it, harmonious in relation to both of these, and with due measure in all other respects. We must therefore, following along these tracks, not practise mathematics in just any way; for the mathematics that is now prevalent makes use rather of sense-perception and imagination, is alien to truth, and has grown up rather as a friend to generation. If, then, we wish to practise mathematics in the Pythagorean manner, it is fitting to pursue with zeal that path of it which is divinely inspired, leads upwards, purifies and brings to perfection.

Chapter 23

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That it was not at random that the Pythagoreans advanced the mathematical sciences further, and indeed beyond the use necessary for life; and what the causes of this are: a reminder at greater length.

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That it was not at random either, then, that Pythagoras transformed the philosophy concerned with the mathematical sciences into the form of a liberal education, and advanced them far both in the multitude of things proved and in the exactness of the demonstrations, and practised them beyond the use necessary for life, is easy to learn from the following. For if we have acquired any seed and beginning of the kind of knowledge from which, having previously received the genus of science by name, we came to see exactly what sort of thing it is in its nature, it has come to us from no other source than from these. Moreover, the power of science too has been made manifest through its own proper arguments in the demonstrations concerned with these things. Further, when we were trusting at random to many of the appearances, understanding of these matters has set us right, making plain how the truth about them really stands. But above all it is in association with these that we first come to partake of a vision that is liberal and befitting philosophers; for what is akin to each is what is like it in nature, and the sovereign end of the free man's activity in the life proper to him has its reference to himself and to nothing else among things outside him[27] [lacuna in the text] belongs to the sciences mentioned, since they are theoretical, and belongs to them first, because the learning of them holds the first rank in respect of the time of life, having no need of the sort of induction that naturally comes about through habituation from particulars; and the philosopher seems (if, as with the other proper desires that have been named from the affection felt towards some genus, we are to attach the name to him too appropriately from

what he feels) to have a longing for a science that is honourable for its own sake, and not on account of anything else that results from it.

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For some of those who wish to advance them, but who say that we ought to undertake the learning of them because training in them is useful for other studies, would not seem to assign them their fitting rank. For the things for the sake of which they urge us to do this are, both in their own nature and in the arguments customarily offered on their behalf, less akin to the truth, and are no match for them in the exactness of their demonstrations. There is a sufficient sign of this: for these we see enduring and trusted alike throughout by those who handle them, whereas of the others we should find altogether few that are such. Now the philosophy concerned with the mathematical sciences has come to our aid towards many both of the things that are necessary for life and of those that already go beyond need and are honourable in themselves. For we should find that not a few of the craftsmen's arts have received assistance from them; and the philosophy of nature, and any other that holds a rank more honourable than this, we should see making use in its own demonstrations of many things which we have come to see by means of the sciences mentioned. Further, by making us akin to what is ordered and to order, it would also provide a certain exhortation towards virtue and all that is fine. Yet it is not only for such assistance that one would cherish their power, but still more for themselves and for their own nature.

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For it is agreed that there are some sciences that are choiceworthy for themselves, and not only for what results from them; and it is the theoretical sciences alone, above all, that can be of this kind, because they have no end other than contemplation. Now the grounds on which we lay it down that one science is more choiceworthy than another are the same as those on which we lay it down that each is itself choiceworthy. We choose one in preference to another either on

account of its exactness or because it contemplates better and more honourable objects; and of these, the former all would concede to us belongs pre-eminently to the mathematical sciences, and the latter would be conceded by all who assign the aforesaid precedence to the first principles, and suppose that the nature of a principle is proper to numbers and lines and their attributes on account of the simplicity of their essence. Further, the objects of contemplation in the heaven, which hold the most honourable and most divine rank among the things sensible to us, are by nature made known through the science of astronomy, which is in fact one of the mathematical sciences; and it would seem absurd and in no way consistent, while declaring that the philosopher is akin to truth, to think that he ought to seek some other fruit from objects of contemplation such as these, which have a share in the highest truth; and, lover of sights as he is, to require him to take up for the sake of something else sciences of this kind, which are concerned with the most common things in nature and the most divine of the things sensible to us, and which, being full of very many and most wonderful sights, possess an exactness that is not fabricated out of empty arguments, but is their own and secure, derived from the nature that underlies them.

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In general, whatever one might require to belong to those sciences that are choiceworthy for themselves, we shall find the mathematical sciences sharing in all of it. For each of them is concerned with a certain nature, and that an eternal one, containing within itself many wonderful sights, in virtue of the order of its proper attributes and in virtue of its remoteness from the conception derived from sensibles. Further, taking the principles of their demonstrations as well known and credible through themselves, they construct their deductions about these things by means of them in such a way that the demonstrations in these fields are a model for those who wish to draw any conclusion with exactness; which is why it would seem fitting for those who hold that the life passed in philosophising is choiceworthy

in itself to hold that the study of the mathematical sciences is proper and akin to philosophy. With good reason, then, on all these grounds, the Pythagoreans honoured the serious study of the mathematical sciences, and co-ordinated it in manifold ways with the study of the cosmos: for instance, comprehending number by calculation from the revolutions and the difference among them; studying what is possible and impossible for the constitution of the cosmos from what is possible and impossible in the mathematical sciences; conceiving the heavenly revolutions, together with their cause, in terms of commensurable numbers; marking off measures of the heaven according to certain mathematical ratios; and in general constructing the predictive science of nature on the basis of the mathematical sciences, and setting the mathematical sciences, like principles, at the head of the other theorems concerning the cosmos.

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From these, indeed, they also procured many demonstrations for matters concerning nature, and they exhort to what is fine and good, to virtue, and, what is greatest, they practise astronomy in a theological manner thanks to the mathematical sciences. So that for all these reasons it was with good cause that they engaged in a wonderfully serious study of them.

Chapter 24

What the customary manner of the Pythagoreans' occupation with the mathematical sciences was, and what their training in the sciences and their elaboration of them.

Next after this it is worth speaking also of the customary manner of the Pythagoreans' occupation with the mathematical sciences. They, then, detached their accounts of the mathematical sciences from sensibles, and by means of them turned discursive thought round to belief in incorporeal essence, and used them to ferry it across to the intelligibles, and considered above all what in them is fashioned in the likeness of the pure forms and the unitary reason-principles. This,

then, was the manner in which they approached the theorems; and, having once removed the science of them from common and public knowledge, they made the imparting of them too, in the same way, a matter of secrecy; they shared the knowledge of them with very few, and if anything was ever divulged to the many, they expiated this as an impiety; which is why they also turned away those outside their fellowship, as being unworthy to partake of them. For Pythagoras held that one should share the philosophy contained in the mathematical sciences not with all, but only with those with whom one would share one's whole life. And to this association he did not admit men at random, nor just anyone, but made trial of them over a long time and turned away the unworthy.

75

And to those outside his fellowship he did not make common the advance that had come about through him, having made the accounts of these things secret towards the others; but among those who were named Pythagoreans on account of their comradeship with him he brought about a great advance both in the philosophy concerned with the mathematical sciences and in the study of geometry, and of nearly all the things that later progressed further one would find that the beginnings came to us from him. What he prized in them was not, as some of those who came later did, the power by which men will be able to discover what has been proposed, but the theorems themselves; and among these not those that were hardest to discover, as most of the later men did, but those in which it was most possible to discern an order among them or some natural property. They were so disposed because they believed that the principles of the whole of nature are present in these things, and that here it is most easily seen what they are and how many, since these things concern a nature that is abiding and free from motion, and moreover simple; which is why they neither touched on problems, except such as were elementary, like the application of areas and squaring, nor, in the theorems, did they labour with the aim of going through everything and leaving out

nothing that is possible, but sought only to see the principles themselves in each case. Their training in these sciences and their rational elaboration of them they made exact and theoretical, directed to a science proper to them; and they added to the sciences the order that befits them;

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and, having received a few things at the outset, they worked these out, and brought to completion above all the most honourable and most august of the theorems; and theorems that were being practised in one way they turned round to other uses; and they established in them an order such that the simpler were handed on first and those involving composition second; and they arranged the theorems in conformity with the nature of the things that are, suitably to our capacity, appropriately to the worth of those receiving them, in agreement with the guidance towards virtue and with education as a whole, and fittingly for the purification of the soul.

76

Such are the marks that one might set down, on this point too, of the Pythagorean type; about which we shall say more in what will be said separately about each of the mathematical sciences.

Chapter 25

Who the mathēmatikoi among the Pythagoreans were and in what respect they differed from the akousmatikoi, and what their work was and of what sort the form of their arguments and demonstrations.

There are two forms of the Italian philosophy, which is called Pythagorean. For there were also two kinds of those who practised it, the akousmatikoi and the mathēmatikoi. Of these, the akousmatikoi were acknowledged to be Pythagoreans by the others, but they for their part did not acknowledge the mathēmatikoi, nor that their system of study was that of Pythagoras, but rather that of Hippasus. Some say that Hippasus was a Crotoniate, others a Metapontine. But

those of the Pythagoreans who are concerned with the mathematical sciences both acknowledge that these are Pythagoreans and say that they themselves are so still more, and that what they themselves say is true.

77

The cause of the unlikeness, they say, was as follows. Pythagoras arrived from Ionia and Samos at the time of the tyranny of Polycrates, when Italy was at its height, and the foremost men in the cities became his intimates. With the older of these, who had no leisure because they were held fast in political affairs, since it was difficult to deal with them by way of the mathematical sciences and demonstrations, he conversed in bare terms, judging that they would be benefited no less by knowing what they ought to do even without the cause, just as those under medical treatment, though they do not hear in addition why each thing must be done by them, none the less attain health. But with all the younger men he met who were able to toil and to learn, with such he dealt by way of demonstration and the mathematical sciences. They themselves, then, so they say, are descended from the latter, and those others from the former. About Hippasus they say that he was one of the Pythagoreans, but that because he was the first to divulge and to draw the sphere composed of the twelve hexagons[28] he perished at sea as one guilty of impiety, and gained the reputation of having discovered it, whereas everything belongs to that man; for so they refer to Pythagoras and do not call him by name. The mathematical sciences advanced, once they had been divulged, two men above all carrying them forward, Theodorus of Cyrene and Hippocrates of Chios.

78

The Pythagoreans say that geometry was divulged in the following way. One of the Pythagoreans lost his property, and when this misfortune had befallen him he was permitted to make money from geometry. Now geometry was called by Pythagoras inquiry (historia). As regards the difference between the two systems of study, then, and

as regards the mathematical sciences, this, more or less, and things of this kind, is what came about. The Pythagoreans, having devoted themselves to the mathematical sciences, and having come to love the exactness of their arguments, because these alone, of the things that men practised, possessed demonstrations, and seeing that the facts of harmony were placed beyond dispute by means of numbers, and equally[29] the mathematical sciences concerned with sight by means of diagrams, supposed that these things and their principles were in general causes of the things that are; so that anyone who wishes to contemplate how the things that are stand must look to these, namely the numbers and the geometrical forms of the things that are, and ratios, because all things are made clear by means of them. Holding, then, that they could attach the powers of each thing to nothing more opportune or more honourable than to the causes of all things and what is primary, they went on to define the other things too in much the same manner as these. Their reductions of things to numbers and the mathematical sciences, then, were made for these reasons and seemed to them to have this pattern. Of some such kind too was their method of demonstrations, setting out from principles of this sort and possessing in this way what is trustworthy and secure in its arguments.

Chapter 26

79

Objections to the mathematical sciences as being of no worth, and replies to them and counter-arguments at greater length.

79

There have been some, both among the ancients and among the moderns, who have put forward the contrary opinion about the mathematical sciences, censuring them as utterly useless and as contributing nothing to human life. Some argue in this way: if the end of them is useless, the end for the sake of which the philosophers say one ought to learn them, then all the more must the serious pursuit of

them be vain. And about the end nearly all are agreed who are reputed to have attained the greatest exactness in that pursuit. For some say that it is the science of things unjust and just and bad and good, a science similar to geometry and the others of that kind, while others say that it is the wisdom concerning nature and truth of that sort, such as Anaxagoras and Parmenides and their followers introduced. Now whoever is going to examine these matters must not fail to notice that all good things, and all things beneficial to human beings for life, consist in use and action, and not in knowing alone; for we are not healthy by being acquainted with the things productive of health, but by applying them to our bodies; nor are we wealthy by knowing wealth, but by possessing much property; nor, most important of all, do we live well by knowing certain of the things that are, but by acting well; for this is truly to be happy. So that it befits philosophy too, if indeed it is beneficial, either to be a doing of good things or to be useful for such actions.

80

Now that neither this nor any other of the sciences mentioned above is any kind of performance of practical affairs is plain to all; and that it is not useful for actions either, one may learn from the following. For we have a very great example in the sciences similar to it and the opinions that fall under them: of the things which the geometers contemplate through demonstration, we see them to be practitioners of none; rather, to divide a piece of land, and all the other properties of magnitudes and of places, the land-surveyors are able to deal with through experience, whereas those who are occupied with the mathematical sciences and the accounts of these things know how one ought to act, but are not able to act. The case is similar with music too and the other sciences in which the part of knowledge and the part of experience are divided and stand apart. For those who have determined the demonstrations and the syllogisms concerning concord and the other things of that kind are accustomed, like those engaged in philosophy, to inquire, but have no share in any of the

works; indeed, even if they happen to be able to do something of them with their hands, as soon as they have learned the demonstrations they at once, as though on purpose, do them worse; whereas those who are ignorant of the accounts, but have been trained and opine correctly, are wholly and entirely superior for practical needs. Likewise also in what concerns astronomy, for instance about the sun and moon and the other stars, those who have studied the causes and the accounts know nothing of what is useful to human beings, whereas those who possess what these people call the nautical sciences are able to foretell to us storms and winds and many of the things that come to pass.

81

So that for actions such sciences will be utterly useless; and if they fall short of right actions, the love of learning falls short of the greatest goods.

81

In reply to these arguments, we say both that there are sciences of mathematics and that these are capable of being acquired. For both are always more knowable: what is prior than what is posterior, and what is better in nature than what is worse. For there is science of things determinate and ordered more than of their contraries, and again of causes more than of their results. Now the things in the unmoved mathematical forms are determinate and ordered. And the prior are causes more than the posterior; for when the former are done away with, the things that have their essence from them are done away with: lengths when numbers are, planes when lengths are, and solids when planes are. So that if indeed the things in the mathematical sciences are simpler than all things, they will also have more of the character of principles than all things. So that there will be sciences far more of the better and more originative things, and they are capable of being acquired; for it is far more necessary that there be wisdom of the causes and the elements than of the things posterior. For these latter are not among the highest things, nor do

the first things arise by nature out of them, but it is out of those and through those that the other things too manifestly come to be and are constituted. And that to know the mathematical sciences is also the greatest of goods and the most beneficial of all the others is clear from the following. For reason and wisdom take the lead among goods; and there is no other rule and most exact standard of goods than the wise man; for whatever things he would choose, these are goods, and the contraries of these are evils.

82

And since all men choose most of all the things that accord with their own proper states (for the just man chooses to live justly, the man who has courage to live in accordance with courage, and the temperate man to be temperate), it is clear that in like manner the wise man too will choose the exercise of wisdom above all things; for this is the work of this faculty. So that it is evident that, according to the most authoritative judgement, wisdom is the best of goods. And it is not for the sake of use that all should pursue it; for it is in fact choiceworthy on its own account.

82

Concerning the benefit and the greatness of the thing I consider that enough has been demonstrated; but that the acquisition of it is far easier than that of the other goods, I am persuaded from the following. For the fact that, though no pay comes to philosophers from men for the sake of which they might labour so intensely, and though they have allowed the other arts a long start, they have nevertheless, running for a short time, passed them by in points of exactness, seems to me to be a sign of the ease that belongs to philosophy. Further, the fact that all are fond of dwelling upon it and wish to devote their leisure to it, letting go of everything else, is no small proof that close application to it comes with pleasure; for no one is willing to toil for a long time. In addition to these points, the exercise of it differs very greatly from all others: for they need neither instruments nor places for their work, but wherever in the inhabited

world one sets one's thought, from every side alike one lays hold of the truth as though it were present.

83

But these things have perhaps been said sufficiently for the present occasion; for it has been demonstrated both that wisdom is possible and that it is the greatest of goods and easy to acquire.

83

Now exact reasoning about the truth is by common consent the youngest of pursuits. For after the destruction and the flood men were compelled first to philosophise about what concerns nourishment and staying alive; then, having become better provided, they worked out the arts that aim at pleasure, such as music and those of that kind; and it was when they had more than the necessities that they then set their hand to philosophy. And those who inquire into geometry and reasonings and the other branches of education have now advanced so far, from small beginnings and in a very short time, as no other class has done in any of the arts. And yet, whereas all join in urging on the other arts, honouring them publicly and giving pay to those who possess them, those who busy themselves with these studies we not only do not encourage but often even hinder; yet all the same these studies make the greatest progress, because by nature they are the eldest; for what is later in generation takes precedence in essence and perfection.

83

And so the science of mathematics too prevails over the other sciences in all these respects by a wide margin, excelling all pursuits in beauty and exactness; and it is also reasonable that this should be so.

84

For, first, the things that are akin in nature to generation are eagerly sought after by human beings, so as to acquire them as far as they are able; but, after these, the things that release us from the corporeal nature are far more honourable than the former; for the former are presupposed as necessary, while the latter, as choiceworthy for their

own sake and august, have been deemed worthy of precedence and honour. Now the mathematical sciences do contribute to us not a few uses for the whole of human life as well, as is plain to those who survey in life the works that come from the mathematical arts; nevertheless these are worthy of less serious attention, and the greatest things are the purification of the immortal soul, and the turning round of the intellect towards the intelligible, and participation in the activity of being. And mathematical science, in procuring these for us, furnishes all goods, so that I do not know whether any other method lends such help towards the end that is happiness. By these arguments, then, not only have the opposing arguments been shown to be false, but the mathematical sciences have also been demonstrated to be most useful to us.

Chapter 27

What the truly educated person ought to demand of the mathematician, and how his study is to be judged, and by what standards its correctness is to be comprehended.

Since it is the work of the educated person to be able to judge aptly what the speaker sets out well or not well, we think that the person educated in general is someone of just this kind, and that to be educated is to be able to do what has been said. It is clear, then, that in the mathematical sciences too the rightly educated person ought to demand of the mathematician correctness and his proper work, as to whether he conducts his study of them well or not well.

85

For just as we consider the person educated without qualification to be, though one in number, capable of judging about practically everything, so too concerning some determinate science there would be another person disposed in the same way as the one described, with respect to a part. It is clear, therefore, that for the study of the mathematical sciences too there must be certain standards of this kind, by reference to which the educated person will accept the

manner of the things proved, apart from the question of how the truth stands, whether thus or otherwise. I mean, for example, whether one ought to take each single theorem of mathematics and make determinations about it by itself, for instance about these particular triangles, or whether one ought to examine the common theorems and the things that belong to all, positing them in respect of something common. For many of the same things belong to many genera that are different from one another, as for instance if one were to make the demonstration in so far as they are triangles, or in so far as they are rectilinear figures in common. For if certain things should belong, the same, to things that differ in species, the demonstration of them ought not to have any difference either. But there are perhaps other things which happen to have the same predicate, yet to differ by a difference in species; for example, the similar is one thing in the case of triangles and another in the case of numbers. And for each of the two one must make demonstrations proper to it. It must therefore be examined when one should study in common, according to genus, and when in the proper manner, according to each thing; for to have made determinations about these matters contributes a great part to mathematical education.

86

Further, one must demand of the mathematician his accounts in accordance with the underlying essence, and that he make the manner of his demonstrations appropriate. Just as, then, we bear with the rhetorician when he argues from what is plausible, so one must demand of the mathematician demonstrations that are necessary. But one must not seek the same necessities everywhere, nor the same exactness alike in all things; rather, just as we divide what belongs to the arts by the underlying materials, not seeking exactness alike in gold and tin and bronze, nor in cork and boxwood and lotus-wood, so in the same way in the theoretical sciences too. For the subjects will at once produce differences, whenever some are simpler and others more in composition, and some wholly unmoved and others in

motion, as for instance the things in numbers and in harmony, or those in geometry and astronomy, and whenever of some the principle is intellect and of others discursive thought, while for some there are also certain slight starting-points from sense-perception, as for the heavenly things. For it is not possible to adduce the same causes, or even similar ones, concerning such things, but as far as the principles differ, so far must the demonstrations differ too; for in each of the sciences the manner is akin. Further, and at a still wider remove than these, there is the fact that some pursue their inquiry possessing principles and others not possessing them; so that here too neither similar causes nor similar accounts are to be set forth. And with a view to these matters it is necessary to know what they have that is the same and what different, and what is the same by analogy, and which kinds of science need more things and in which kinds the difficulties are more numerous; for it is roughly in these and suchlike respects that the variations in the demonstrations and accounts concerning each thing lie.

87

And such a study would contribute not only to judging but also to inquiring as one ought; for, possessing a determination of the cause in each case, it will produce appropriate accounts, which is not easy for one who has not been habituated to do. For nature itself by itself is able to lead the way to the principles, but is not sufficient to judge each thing unless it has acquired another understanding in addition. Further, it must be discerned whether there are several causes concerning the things about which the mathematician must speak, and which of these is by nature first and which second. For the person educated mathematically is apt to scrutinise the causes assigned as well, and to study their order.

87

Nor must this escape notice, that many of the more recent Pythagoreans supposed that the only things that remain the same and in the same state are the subjects of the mathematical sciences, and

posited these alone as principles; and so they marked off the sciences too in the same way, as concerned with such things, and the demonstrations as well. But since we shall demonstrate, both in the discussions that now come first and in those that will be stated later, that there are many other essences too that are unmoved and remain the same, not only those of the mathematical sciences, and that those are elder and more honourable than these, and shall also demonstrate that not only are these mathematical ones principles, but there are others as well, and that those indeed are the elder and more powerful than these, and that the mathematical principles are principles not of all the things that are but of some: for these reasons mathematical demonstration now requires a determination, of what sort of things that remain the same and in the same state it is demonstrative, and from what sort of principles it syllogises, and about what sort of problems it makes its demonstrations.

88

For the education that passes judgement on these matters marks off both the correctness and the end of mathematics, makes its judgement of it as is fitting, and comprehends well the manner in which the inquiries ought to be conducted. So let these be our determinations on these matters too.

Chapter 28

When the problem or the manner of the demonstrations belongs to mathematics and when to another science: a scientific distinction.

Since it is often not evident whether the inquiry in hand is mathematical or belongs to another science, and since it is disputed what manner of demonstrations is mathematical, it is necessary to make clear about these matters too what sort of problems, and demonstrated in what way, display a mathematical method. In general, then, one must know beforehand that theological science and physical science lie alongside mathematical theory, so that these sciences share both their demonstrations and their problems with one

another. Now the truth does indeed stand thus: the kinship of knowledge joins together a certain single affinity among the different cognitive faculties, and the things too, being in a way close to one another, possess the common continuity of the sciences with one another. Nevertheless the Pythagoreans, for their part, knit this communion together still more closely, as undivided, handing down many things about the intelligible forms by means of the mathematical sciences, and many about nature, and expounding many things about character too in a mathematical manner. But all the same one must distinguish the three genera of these accounts and determine the mathematical one separately, so that it is not confused with any of them.

89

Let mathematical accounts, then, be unmoved in themselves and irrefutable, fitted to the mathematical forms and genera in themselves, not comprehending these by abstraction from sensibles, but applying themselves to them in a determinate way, inasmuch as these subsist by themselves, neither having any contact with motion, nor coming together into identity with the intelligible and indivisible forms or with the intellections, but proceeding step by step in accordance with the intelligibles and the acts of discursive thought that are constituted about them, and employing this as a certain intermediate mode of knowledge. For a method of accounts and demonstrations of this kind is both mathematical and far separated from the accounts found among the other practitioners of science.

Chapter 29

On mathematical syllogisms and mathematical divisions and definitions: how mathematical science uses them, whether in its own proper manner or by taking its principles from dialectic.

It follows upon problems of this kind to examine this also along with them: how mathematical science uses division, definition and syllogisms, whether as receiving the learning of them from dialectic,

or as itself exercising its activity upon them from itself. Now if it takes these over from the study concerned with reasoning, it will be in want of many things and gets the principles of its own proper knowledge from elsewhere; but if such a thing befits some other science rather than mathematical exactness, we must conceive these three, namely division, definition and syllogism, as being one thing in dialectic and another in mathematics, and as determined in each of the two according to the genus proper to each. Now those of dialectic are greater objects of study, and it is not our present purpose to go through them, whereas those proper to mathematics are the concern of mathematics itself alone.

90

From itself, therefore, it discovers them, perfects them and works them out; it knows well how to test what is proper to itself, and it has no need of another science for its own proper study. For it discerns not what is without qualification, as dialectic does, but the things that fall under itself; it studies them in its own proper manner, in so far as they are subject to it, and about them it makes its exact determinations; it has within itself criteria by which they must be tested; it devises several manners of demonstration, and among these it distinguishes those that are better and true and those that are ambiguous and in need of further attention. Moreover, it makes its treatment twofold, the one directed to discovery, the other to judgement. But it is capable of judgement and of discovery according to the reason peculiar to its own art, not according to the intellect that is contemplative without qualification. It is able, then, in accordance with this reason to discern how one ought to divide the species in mathematics, and what the differences are that properly divide them, and what the definitions in mathematics are, and how one ought to gather them from the mathematical divisions, and how a mathematical syllogism comes about, and by how many determinations it acquires its exactness, and when it is possible and when impossible, and in how many ways it possesses necessity. For in

all these matters its own proper discovery, use and judgement need no preparation brought in from outside, as some suppose.

Chapter 30

That mathematics makes great contributions to the whole of philosophy and to all its parts, ministering to it in all things, and above all the mathematics of the Pythagoreans, which differs greatly from the rest of mathematics.

That mathematics also contributes many and great services to the whole of philosophy and to all its parts, ministering to the vision of the things that are and following along in its track, is easy to learn from what follows.

91

The mathematics of the Pythagoreans is not such as that which the many practise. For the latter is for the most part technical, has no single goal, and does not aim at the beautiful and good, whereas that of the Pythagoreans is pre-eminently contemplative; it refers its theorems to a single end, makes all its own accounts akin to the beautiful and good, and uses them as leading up towards being. Setting out, then, from such a starting-point, it distinguishes well within itself which of the studies belonging to it are fitted to theology, being capable of sharing in divine order and divine measures, and these it assigns to that part of philosophy; and which are appropriate to the hunt for being, with a view to affinity, commensurability and the turning round, and such theorems too it duly renders to the corresponding part. Nor does it escape its notice if some lend their aid in a scientific way to exactness of reasoning, leading the way to syllogising, demonstrating and defining well, refuting what is false and distinguishing what is true from what is false. Nor indeed is it ignorant of the harmony that is appropriate to the inquiry into nature, how it is constituted and how it is useful, how it fills up what is lacking in nature, and how it passes judgement upon these things. Further still, it descends to constitutions, the formation of characters and rightness of life, and discovers the appropriate standards of the

mathematical sciences for both household and cities, and uses them as is fitting for the sake of what is best, both for correction and for the best education, for the good measure that is appropriate, and for guarding against what is shameful and acquiring what is fine, lending its aid in this way both in common, through all the mathematical sciences, and in the proper manner through each of them.

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And moreover, with regard to the goods that accord with nature and to the advantages of the arts, by discovering some things, by bringing others to light as by-products and helping to establish them in the role of an addition, by furnishing works from itself and contributing parts to some, it has perfected human life, so that this is sufficient to itself and lacks nothing of what life requires. Hence we may reasonably fit the mathematics practised by the Pythagoreans together with philosophy, as being akin and well suited to it.

Chapter 31

That the Pythagoreans used the same mathematical sciences for many different things, and made several mathematical sciences indicative of the same thing, and for what reasons.

Since it is of such a kind, we find it employing along many paths the science concerning things. For it often uses the same mathematical sciences for many things, whether physical or theological, or for generation, or for the elements, or for whatever pertains to the production of generation, or for composite things, or for simple ones; for example, it uses numbers for all of these, and harmonies, and figures, and the other things of that kind; and sometimes it takes up together several at once as indicative of the same thing, for example, of the soul: numbers, harmonies, figures, and certain other mathematical objects. The cause of the former is that the natures of each of the mathematical sciences and of each mathematical essence are multiform (for this is why we use the same ones for many things, for example numbers for theological and for physical matters), while

the cause of the latter is that some of the things that are have been blended from many essences and are intermediate between several extremes.

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For this is why the soul and all the intermediate natures are expounded through several mathematical sciences, as being able to fit several mathematical essences and as receiving from several the confirmation of their being. These matters, then, may be taken to admit of some such account.

Chapter 32

How we sometimes treat even of sensibles mathematically, and in how many ways this comes about, and how many things in the mathematical sciences are referred to other things, and for what reasons.

It is the custom of mathematical theory sometimes to treat of sensibles too in a mathematical way, for example of the four elements geometrically or arithmetically or harmonically, and of the rest likewise. For since mathematical theory is prior by nature and sets out from things prior to those that exist by nature, for this reason it also constructs its syllogisms as demonstrative from prior causes. And it does this in several ways: either by abstraction, when, having removed the enmattered forms from matter, it examines them mathematically; or by application, when it brings the mathematical reason-principles to bear upon physical things and joins them to these; or by completion, when, by adding what is lacking, it fills out the corporeal forms, which are incomplete; or by representation, when it looks to see in what respect especially the equal and commensurable things in generation have been made like the mathematical forms; or by participation, when we examine in what respect the reason-principles that exist in other things participate in the pure reason-principles; or by reflection, whenever we observe a faint trace of the mathematical reflected in sensibles; or by division, when we apprehend the one and indivisible mathematical form as it is divided about the particulars

and multiplied; or by comparison, when we examine side by side the pure forms of the mathematical sciences and those in sensibles; or by the cause that derives from prior things, when, having set up mathematical objects as causes, we examine together how the things in sensibles come to be from them.

94

For it is in this way, I think, that we treat mathematically of all the things in nature and in generation. And it is from this cause that many of the things in the mathematical sciences do not remain among the mathematical sciences, but are drawn to things inferior to them, either through the choice of those who use them, or also through the kinship which sensible things have with them. Sometimes, however, the starting-point taken from the mathematical sciences is also referred upwards to greater things, and to all other things, since incorporeal essence is naturally well suited to become akin to the pure essences of the things that are, and because the mathematical sciences are of a nature to be made like all things. So much, then, about these matters too.

Chapter 33

What is common in the whole of mathematical science and what is proper to it, according to the differences observed in many species, and how one must cut it according to the science of division, from one into two and then into more species.

Since the common genus of the whole of mathematical science is of the greatest importance for scientific knowledge of the present study, we must above all discern this: in what respect the whole of mathematics possesses what is common. For if we discern this one genus of it and come to know its common essence, we shall have the most perfect knowledge of it. Now we have already examined this beforehand at the beginning too, but we must also set upon the end a crown that is the same as the beginning which was laid down as foundation.

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So now too let us take up again the account of the one genus of mathematical theory.

95

I say, then, that what is, simply so, intermediate between the intelligible and the sensible forms is the common genus of this science, containing within itself all the different species, as many and of whatever kind they are, participating primarily in the genera of being, having gathered up within itself the genera of sensibles, yet altogether surpassing these in purity, exactness, subtlety and incorporeality, and containing within itself powers of every kind, some leading up to the things that are and others inclining towards generation, and cognitions likewise. Positing this, such as it is, as one, let us conceive its differences according to the dichotomies of this intermediate nature, of which let us mark off the one as holding to being, and the other as laying hold of sensibles and standing before them; and again, taking for each of these two species different terms in turn, we shall grasp the intermediate character of the mathematical sciences in different ways, distinguishing it either according to quantity or according to some other genus of it. The distance from being, too, according as it is further off or near, will furnish the mathematical sciences with their difference. Further, the precedence among the things in the constitution of the universe, according as it is first or second, itself also produces certain differences among the mathematical sciences. And their order we ought to discover either according to succession, or according to continuity, or according to contiguity, or according to some other kind of connection.

96

By this constitution that proceeds from one and is multiplied, and by the ordering that is led back to one, we distinguish, and in distinguishing we likewise join together. And in this way the common essence of the mathematical sciences will be comprehended by us in common by reasoning, and divided, and led back again from the many

to the one genus of mathematical essence; and this indeed is the end of the mathematical science of division and definition.

Chapter 34

Whence the science of mathematics has its name and what its character is, and to what one must attend in judging the form of the mathematical sciences.

The science of mathematics has been so named because a certain learning (mathēsis) descends into it from the intellection of the intelligibles, and this study is akin to that intellection, and learning arises about just those things with which intellection and science are concerned, and these alone among all things would be learnable (mathēta), which is why they have been called mathēmata. The Pythagoreans held that these mathematical sciences had been provided as, so to speak, proper instruments for unfolding the nature of the things that are and for removing all the mist that darkens things, so that one might behold the truth itself in its purity; and further that such learning would also help to supply the cause of the agreement in matters of character, and, becoming an ally to philosophy, would help to unveil the nature of good and evil, and that through it the commensurability of the order of the cosmos and of the circular motion of the heaven is contemplated. For this reason they thought that it was not possible to philosophise without it. Further still, it hunts down, as though through mirrors of surpassing brightness, many images of the works of nature, and this they used to name the mathematical part of philosophy, and those experienced in accounts of this sort they declared to be mathēmatikoi.

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They also held that the things in the mathematical sciences are the finest paradigms of the things here, because they are regulated with reference to intellection, being stable and unmoved and always remaining the same and in the same state, so that by looking to them and imitating them one would produce in each case works that are

steady and secure. The mathematical account possesses, moreover, purity, scientific character and irrefutability. And to the ancients the mathematical account seems to be prior to the physical and more originative; for on it, they supposed[30], the pure intellection of the other things depends. This, then, is the character of the mathematical account. And in judging its form one must attend to its exactness, its irrefutable knowledge, the correctness of its accounts and its agreement with the things that are; for it is together with these that its form shows through best. Let these too, then, be our additions to what was determined about these matters earlier.

Chapter 35

Recapitulation of the common account concerning all the mathematical sciences, an exhibition of the order of the chief heads, and at the same time a reminder that the whole synopsis of them has been correctly divided.

Since, then, we have completed the common theory of the mathematical sciences, it is now time to gather under a single synopsis all the chief heads concerning them. First, then, we gave an account of what concerns the aims and of what concerns mathematical science as a whole, then of principles, both the common and the proper, and after these of the things that underlie the mathematical sciences;

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then of their best use and of the objects of science in them, and of the criteria that are theirs in themselves, and of their determinate essence, in how many ways it is considered; and what the work of mathematical theory is, and what its powers are, and what its elements, both the common and the proper; and how all these have communion with philosophy, and how much they contribute to the arts, and what order they have in the training that leads to education; and what the proper modes are of the Pythagorean transmission of the mathematical sciences, and what the division is, according to the Pythagoreans, of the whole of mathematical science, and what

definitional mathematics is, and what the special features of this study are according to Pythagoras, and what his own peculiar practice of it was, and that it was not at random that the Pythagoreans advanced these sciences to the furthest point; and what their customary manner was of occupying themselves with the mathematical sciences, and who the mathēmatikoi were among them; objections to the mathematical sciences, and the replies to these and the counter-arguments; and what the truly educated person ought to demand of the mathematician; and the distinction of the problems in it and of the manner of the demonstrations; on mathematical syllogisms, divisions and definitions; and what communion philosophy has with the Pythagorean mathematics which the Pythagoreans employed in the mathematical sciences; and what is common and what proper in mathematical science, and how many divisions it has and how they are ordered; and at the end the name of mathematics has been discussed, and what follows upon this. So many are the chief heads that we have examined concerning the common account of mathematics and of the mathematical sciences, and I think that they are in due measure.

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And if any have anywhere been left out, these too will be easy to discern from what has been said.

Notes

[1] συνεπιφέρει μὲν οὐ συνεπιφέρεται δέ is translated as it stands; of what is prior by nature the converse is expected (συνεπιφέρεται μέν, οὐ συνεπιφέρει δέ).

[2] The clause εἰ παντάπασιν εἴη διήκων κατὰ ταῦτα τὰ παραδείγματα is obscure and perhaps corrupt; it is rendered by its apparent sense.

[3] καὶ ὅσον is read for ποσόν (after τοῦ ποσοῦ), which cannot be construed; the next clause begins in the same way (καὶ ὅσον δὲ αὖ).

[4] Iamblichus' text has ἐμβλέπων where Epinomis 991e has εἰς ἓν βλέπων ('looking to one thing', ὃ λέγομεν being then a parenthesis); it is translated as it stands.

[5] Iamblichus' text has τὸ καθ' ἕν τε καὶ κατ' εἴδη where Epinomis 991c has τὸ καθ' ἓν τῷ κατ' εἴδη ('the particular must be brought under the species'); it is translated as it stands.

[6] Iamblichus' text has ἡ ὄψις αὐτή ('sight itself') where Republic 523d has ἡ ὄψις αὐτῇ ('sight ... to it', that is, to the soul); it is translated as it stands.

[7] ἀχώριστα is read for χωριστά, with Republic 524b–c; the word as it stands gives the opposite of the sense required.

[8] ὁρώμενα is read for δρώμενα ('done'), with Republic 530b.

[9] ὁρατῷ is read for ἀοράτῳ ('invisible'), with Republic 532c–d; the region meant is the bodily one.

[10] The verb ἀπαιδευτῇ gives no satisfactory sense with τὸ Ἀρχύτειον as its subject and is rendered by its apparent sense.

[11] καταλήγοντα (with ἐχόμενα) is read for καταλήγουσα, which has nothing to agree with.

[12] Iamblichus' text has παίζων alone where Epinomis 992b has παίζων καὶ σπουδάζων ἅμα ('in play and in earnest at once'); it is translated as it stands.

[13] In ἐπιστάμαν γὰρ καὶ λόγον τὸν προσείληφε the word τὸν cannot be construed and is left untranslated.

[14] The clause ταῦτα δὲ σύμφωνα ποιητά, θεωρούμενα δι' αὐτῶν ἀλάθεα is probably corrupt and is rendered by its apparent sense, ποιητά being taken as ποιητέα and αὐτῶν as αὑτῶν.

[15] The text has ἴσα ('equal') where the parallel in Republic 509d has ἄνισα ('unequal'); it is translated as it stands, and Iamblichus' own paraphrase (ἴσας δὲ αὐτὰς τίθεται) presupposes it.

[16] ποττἆλλα is translated as it stands; ποτ' ἄλλαλα ('relative to one another') is expected, as in Republic 509d and in Iamblichus' paraphrase (πρὸς ἄλληλα).

[17] The text has ὡς εἰκός where Republic 510d–e has ὡς εἰκόσιν ('as likenesses'), which the argument requires; it is translated as it stands.

[18] The sentence is translated as it stands. The negative of Republic 511b (οὐκ ἀρχὰς ἀλλὰ τῷ ὄντι ὑποθέσεις) is absent, so that the hypotheses are called at once hypotheses and ἀρχάς.

[19] προγνωστικόν is translated as it stands; γνωστικόν ('capable of knowing') is expected.

[20] ἀλόγων (‘of things irrational’) is translated as it stands, though it sits oddly with συμμετρία; λόγων or ἀναλόγων may be expected.

[21] δυναμένας is read for δυναμένων, in agreement with τὰς σταθερὰς … θεωρητικάς.

[22] σύντασιν (‘concerted effort’) is translated as it stands; σύνταξιν or σύστασιν (‘system’, ‘constitution’) may be expected.

[23] τὸ ἀδιάφορον (‘what is without difference’) is translated as it stands; τὸ διάφορον is expected, since the reason that follows is a reason for a difference.

[24] σπαραττομένοις (‘torn asunder’) is translated as it stands; παραγομένοις (‘produced’) is expected of the effects of a cause.

[25] τυγχάνει ὂν τὸ ὑψηλόνουν is read for τυγχάνει τὸ ὂν τὸ ὑψηλόνουν, which cannot be construed.

[26] καθ’ ἑαυτήν is translated as it stands and referred to the soul; καθ’ ἑαυτάς (‘existing by themselves’) may be expected.

[27] A lacuna is marked in the text after τῶν ἐκτός; the subject of ὑπάρχει is lost with it, and the English breaks off at the same point.

[28] ἑξαγώνων (‘hexagons’) is translated as it stands. The faces of the dodecahedron are pentagons, and πενταγώνων is expected, as in the parallel account in De vita Pythagorica 88.

[29] ἔνισον is not a recognisable form and is rendered by its apparent sense (‘equally’).

[30] ὑπελάμβανον is read for ὑπελάμβανε, which has no singular subject after τοῖς παλαιοῖς.